



**Sentiment Analysis of Social Media Post on
#BlackLivesMatter Political Movement Women
in Twitter Platform**

Student Name: DHARANI MANIKANDAN

M.Sc. Data Science

October 2024

Supervisor: Mohammed Saeed

School of Computer and Engineering Sciences

Faculty of Science, Business and Enterprise

SignedDHARANI

MANIKANDAN..... (student)

Date03.10.2024.....

ACKNOWLEDGMENT

Completing this research project is one of the most significant achievements in my educational life. I would not have achieved this specific milestone without the support and input of several people. This section acknowledges these people and expresses my gratitude. First, I would like to thank my supervisor, **Mohammed Saeed**. Their immense experience and knowledge were instrumental in guiding me towards the right direction during the journey. Additionally, their timely advice helped this research overcome several hurdles. Finally, I would like to thank my closest friends and immediate family. They provided me immense material and moral support with things such as study material and encouragement. This research would not have been successful without you.

ABSTRACT

Social media has become the new “town square”. Sites such as X (referred to as Twitter in this report), have highly influential users who constantly update the public on various information. Additionally, significant political events are discussed on the forum. This project wanted to create an automated way to measure public sentiment on political events using sentiment analysis. The study used Selenium to gather tweets related to the Black Lives Matter movement using its hashtag. The data was cleaned and explored using visualisations. Next, the textual data was converted into a numerical form using techniques such as tokenization, lemmatisation, and stop-word removal, before training several models. These models are Decision Tree, Random Forest, BERT, XGBoost, and Logistic Regression. The Random Forest algorithm produced the best model with 59% accuracy. The other results included 58% accuracy for the XGBOOST model, 52% for the deep learning BERT model, and finally 50% accuracy for the baseline Logistic Regression model. The relatively average performance shows that sentiment analysis on Twitter is a non-trivial task that requires more complex modelling.

Table of Contents

ACKNOWLEDGMENT.....	iii
ABSTRACT.....	iv
1 INTRODUCTION	1
1.1 Background of the study	1
1.1 Motivation of the study	2
1.2 Research Aim	2
1.3 Research Objectives	2
1.4 Research Question.....	3
1.5 Research Hypotheses.....	3
1.6 Significance of Study	3
1.7 Dissertation Structure	4
2 LITERATURE REVIEW.....	5
2.1 Introduction.....	5
2.2 Introduction to #BlackLivesMatter movement	5
2.2.1 History and goals of the movement	5
2.2.2 Role of social media in the movement.....	5
2.2.3 2.1.3 Participation of women in the movement	6
2.2.4 Key issues and campaigns	7
2.3 Sentiment analysis methods	7
2.3.1 Lexicon-Based Methods	7
2.3.2 Machine Learning Methods	8
2.3.3 Deep Learning Methods.....	9
2.3.4 Comparison of Different Methods	10
2.4 Prior studies on Sentiment Analysis of Social Movements	11
2.4.1 Analysis of #BlackLivesMatter on Twitter.....	11
2.4.2 Analysis of other social movements on Twitter.....	12

2.4.3	Gender differences in sentiment and participation	12
2.4.4	Challenges and limitations of prior studies.....	12
2.5	Summary	12
3	RESEARCH METHODOLOGY.....	14
3.1	Introduction	14
3.2	Research Philosophy	14
3.3	Research Approach.....	14
3.4	Data Collection and Analysis	15
3.5	Research Design.....	15
3.6	Research Strategy	15
3.7	Time horizon	16
3.8	Development Methodology.....	16
3.9	Ethical Considerations.....	17
3.10	Chapter Summary	18
4	IMPLEMENTATION.....	19
4.1	Introduction	19
4.2	Tools used.....	19
4.3	Data Collection and Understanding	19
4.3.1	Data Collection Procedure	19
4.3.2	Collected Dataset Information	20
4.4	Data Pre-processing.....	21
4.4.1	Data Cleaning.....	21
4.5	Feature Engineering	22
4.5.1	Feature extraction.....	22
4.5.2	Data Transformation	23
4.5.3	Data Analysis	23
4.5.4	Textual Analysis.....	24

4.5.5	Twitter Metrics Analysis	27
4.6	Modelling	32
4.6.1	Logistic Regression.....	32
4.6.2	XGBoost (Extreme Gradient Boosting).....	33
4.6.3	BERT (Bidirectional Encoder Representations from Transformers)	33
4.6.4	Random Forest	34
4.6.5	Decision Tree	34
4.6.6	Data Vectorization.....	34
4.7	Evaluation metrics.....	34
4.8	Chapter Summary.....	35
5	RESULTS AND DISCUSSION.....	36
5.1	Introduction	36
5.2	Model Performance	36
5.2.1	Logistic Regression.....	36
5.2.2	XGBoost	37
5.2.3	BERT.....	38
5.2.4	Random Forest	40
5.2.5	Decision Tree	42
5.3	ROC Curve Analysis for Sentiment Analysis Models	45
5.4	Comparative Analysis	46
5.4.1	Validation Analysis for Models (User test Analysis)	46
5.5	Validation Process for Sentiment Analysis Models (User Test).....	47
5.6	Conclusion.....	47
6	ANALYSIS OF RESULTS	48
6.1	Introduction.	48
6.2	Key Findings	48
6.2.1	Key findings from the EDA	48

6.2.2	Key Findings from the Modelling Phase	48
6.3	Implications of the research.	48
6.4	Limitations of the research.	49
6.4.1	Informal Language:.....	49
6.4.2	Sarcasm and Irony:	49
6.4.3	Subjectivity	49
6.5	Conclusion.....	49
7	CONCLUSIONS.....	50
7.1	Introduction	50
7.2	Conclusion.....	50
7.3	Achievement of Objectives	50
7.3.1	Objective 1: Review and Critically Evaluate Existing Literature on Sentiment Analysis Methods.....	50
7.3.2	Objective 2: Data Collection and Pre-processing	50
7.3.3	Objective 3: Visualization of Tweet Engagement Metrics.....	51
7.3.4	Objective 4: Design, Develop, and Implement Machine Learning Models	51
7.3.5	Objective 5: Evaluate ML Model Performance	51
7.4	Recommendations	51
7.4.1	Data Diversification and Volume	51
7.4.2	Model Optimization	51
7.4.3	Handling Informal Language and Emojis.....	52
7.5	Conclusion.....	52
8	FUTURE WORK.....	53
8.1	Introduction	53
8.2	Incorporating Emojis in Sentiment Analysis.....	53
8.3	Optimization Techniques.....	53
8.4	Multimodal Sentiment Analysis	53

8.5	Longitudinal Analysis	53
9	REFERENCES	54
10	APPENDIX.....	58
10.1	Appendix A: Setting up Login.....	58
10.2	Appendix B: Obtaining Tweets from the #BlackLivesMatter hashtag.....	58
10.3	Appendix C: Collecting the tweets from the hashtag.	59
10.4	Appendix D: Data cleaning and text preprocessing	60
10.5	Appendix E: Creating and visualizing sentiment labels	61
10.6	Appendix F: Logistic Regression Modelling.....	61
10.7	Appendix G: XGBOOST Modelling	63
10.8	Appendix H: BERT Modelling.....	63
10.9	Appendix I: Random Forest Modelling.....	66
10.10	Appendix J: Ethics Application Form	66
10.11	Appendix K: Data set(Link).....	67

Table of Tables

Table 5.1	Performance metrics for the Logistic Regression.	36
Table 5.2	Performance metrics for this model.	38
Table 5.3	Performance metrics by BERT model.....	39
Table 5.4:	Random Forest representing multiple trees and merges.	40
Table 5.5	Decision tree representing the most significant attribute.	42

Table of Figures

Figure 2.1:A lexicon-based approach for sentiment analysis of multimodal content in tweets Source: (Nartey, 2022).....	7
Figure 2.2:How Twitter use machine learning Source: (Li et al., 2021).....	9
Figure 2.3:Deep Learning Methods tweets Source: (Lampinen, 2020).....	10
Figure 4.1 Twitter log in page.....	20
Figure 4.2tweets requested directly from twitter using selenium.	20
Figure 4.3 Collected data set attributes.....	21
Figure 4.4: Distribution of sentiment labels	22
Figure 4.5New data set after pre-processing.....	23
Figure 4.6 Quantitative summary of the dataset.	24
Figure 4.7 Quick visual understanding of themes presented in the tweets.	24
Figure 4.8 Histogram of Tweet's overall distribution.	25
Figure 4.9 Box plot comparing word length of tweets across different sentiment categories.	26
Figure 4.10 Line Plot of tweets posted each day from the organisation.....	27
Figure 4.11 Correlation Matrix of likes, Retweets, and Replies.....	28
Figure 4.12 Visualizations an hourly distribution of tweets ,likes, retweets and replies.	29
Figure 4.13 Histogram illustrating analysis of engagement metrics.....	30
Figure 4.14 Barb chart displaying top 10 hashtags.....	31
Figure 4.15 Pie chart displaying top 10 mentions in the tweets	32
Figure 5.1 Classification Report for the Logistic Regression Model.	37
Figure 5.2 Classification Report Plot Matrix for the XGBoost Model.....	38
Figure 5.3 line graph representing training and validation loss for model BERT.	39
Figure 5.4 Classification Report Plot Matrix for the Random Forest Model.	42
Figure 5.5 Classification Report Plot Matrix for the Decision Tree Model.....	44
Figure 5.6 ROC Curve Analysis for Sentiment Analysis Models.....	45
Figure 5.7: Comparative Analysis of Model Accuracies	46

1 INTRODUCTION

1.1 Background of the study

The Black Lives Matter movement started in 2013 following the acquittal of George Zimmerman in the fatal shooting of an African American teenager, Trayvon Martin (Ahaotu and Oshamo, 2023). Since then, it has grown in the ways of visibility and demand for non-violent civil obedience and civil protest rights activism to combat cases of police brutality and all racially motivated violence against women (Bowman et al., 2021). It uses social activism, protests, and advocacies through social media platforms, including the use of the hashtags #BlackLivesMatter, to protest against prejudice and the discrimination of black Americans (Brünker et al., 2020).

Researchers have begun to explore the role, effects, and communication practices of social media in contemporary social movements. There are other textual analysis studies that have been done on the tweets that contain #BlackLivesMatter to identify how the movement situates issues and events, calls for support and shapes opinions (Ahaotu and Oshamo 2023; Alfano et al. 2023). Other similar analyses have indicated that the movement is positive as it encompasses assertiveness, affirmation of power and voice, representation and desensitization, and nonacceptance of stereotyping (Alfano et al., 2023). Several studies have employed data mining and natural language processing (NLP) techniques to analyse extensive Twitter datasets, effectively mapping information diffusion within social media networks associated with specific movements. These analyses not only track how information spreads among users but also evaluate the sentiment and emotional tone of messages shared on the platform. By examining the perspectives of various stakeholders, researchers can gain insights into public opinion and engagement regarding the movement, thereby enhancing our understanding of social media's role in activism (Badaoui, 2020).

Some limitations remain concerning the possibility to develop a comprehensive understanding of the differences in participation and perceptions based on demographic characteristics, including gender. While the #BlackLivesMatter movement has incorporated both male and female participants on social media platforms, there is a lack of scholarly research on the involvement of women in the movement and their contribution to the constitution of the movement's discursive practices and activities. It becomes easier to treat some of the gender-

specific concerns and get the view of the other gender on power relations and representation issues within the movement.

Advanced algorithms from the field of machine learning and natural language processing will be applied to the task of automatic classification of tweets into positive, negative or neutral. This will offer important information about how women have cared, commented and campaigned for change on the issues through their online presence and conversations on social media. It may provide a more comprehensive understanding of the movement and evaluate the progress toward including intersectional feminism in this fight for race and gender equity.

1.1 Motivation of the study

This study aims to integrate a gender-focused perspective that has been overlooked in prior research on the Black Lives Matter movement, specifically in relation to its digital activism and discursive practices on social media platforms. Although previous research has provided useful information, the majority of the works have been limited to exploring general patterns of participation with no regard to how women may have had different ways of engaging in the conversations depending on intersectional feminism (Chan et al., 2023). This research seeks to fulfil this gap by analyzing the sentiments of women on the Twitter platform regarding the movement. The knowledge to be gained will improve comprehension of power relations and representation in the context of the movement from a gender perspective. It is important to have such a perspective to evaluate the current initiatives to promote multicultural representation and intersectional justice.

1.2 Research Aim

This dissertation aims to perform a sentiment analysis of Twitter posts about the #BlackLivesMatter movement, focusing on the emotional tone of conversations, particularly highlighting the experiences and perspectives of women.

1.3 Research Objectives

1. To develop a thorough understanding of sentiment analysis techniques by reviewing and critically evaluating existing literature on sentiment analysis methods
2. To collect a diverse dataset of social media posts from Twitter platforms and pre-process the data to remove noise, slang, and irregular text formats

3. To visualize the frequency and patterns of tweet engagement metrics (likes, retweets, replies) over time, identifying key trends and periods of heightened activity.
4. To design, develop, and implement machine learning models for sentiment analysis, considering the unique characteristics of social media language.
5. Evaluate ML model performance by considering factors such as accuracy, precision, recall, and F1 score.

1.4 Research Question

What emotions do women express on Twitter regarding the #BlackLivesMatter political movement, particularly concerning their experiences with police violence and racial discrimination?

1.5 Research Hypotheses

1. Null hypothesis (H_0): There is no significant difference in sentiment expressed by women on Twitter regarding the #BlackLivesMatter political movement.
2. Alternative hypothesis (H_1): There is a significant difference in sentiment expressed by women on Twitter regarding the #BlackLivesMatter political movement compared to sentiments expressed by men on the same platform.

1.6 Significance of Study

This study is addressing the current gaps in research around analyzing women's experiences within the Black Lives Matter movement on social media platforms. Using new methods of machine learning and natural language processing, the study will offer fresh quantitative evidence exploring how women have various attitudes towards the movement over time. The knowledge that would have been obtained would have provided a better representation of marginalised voices and perspectives that are often sidelined. This shift in focus was necessary in order to evaluate current progress towards diversification and representation adequately. More generally, the insights will be useful in furthering the analysis of shifting power relations within contemporary social justice struggles in the context of digital media. The proposed research is timely providing fresh data for policymakers and advocates to consider when assessing the progress made and the gaps that need to be addressed from an intersectional feminist perspective.

1.7 Dissertation Structure

This dissertation begins with the introduction chapter which aims to introduce the topic. It gives the background, the motivations for the research, and the research objectives. The second chapter is the literature review. This section gives the theoretical foundations of the research. It will sample previous scientific literature on the topic to identify the state of the art. Additionally, the review will aim to identify important contributions and research gaps that can be exploited. The third chapter is the research methodology. This section provides the plan that this research will use to conduct and assess its experiment. The fourth chapter will provide details about the implementation of the research methodology including tools used, algorithms used, and references to the code. The next chapter will be the results chapter. This chapter will state the results achieved in the experiment. The sixth chapter is the analysis, which discusses the meaning of these results and their implications on the state of the art. The seventh chapter gives the final remarks on the project. These include conclusions, achievement of objectives, and limitations. The final chapter is the future work. This chapter will briefly mention plans to improve on the limitations and opportunities for this research.

2 LITERATURE REVIEW

2.1 Introduction

This second chapter of the dissertation aims to give a critical analysis of several research papers in the field of social justice and machine learning. First, the review introduced the political movement that this research hypothesizes will drive public opinion. The second part of the review covered the different types of sentiment analysis methods, including machine learning and deep learning. Finally, the chapter covered previous work in sentiment analysis.

2.2 Introduction to #BlackLivesMatter movement

2.2.1 History and goals of the movement

The Black Lives Matter movement was founded in 2013 as a reaction to the acquittal of George Zimmerman for the shooting death of an African American youth, Trayvon Martin (Field et al., 2022). It emerged from concerns over what some people believed was ongoing police misconduct and racism in the American criminal justice system (Dunklin and Jennings, 2022). The main objectives of the movement were to promote nonviolent civil resistance and public demonstrations to increase public awareness and force change in policies that require it. Some of the movement's requirements included the elimination of racial profiling, reducing the militarization of local police departments, mandatory implicit bias training, and creating independent review boards to investigate cases of police brutality (Field et al., 2022). In the subsequent years, the movement broadened its focus to other matters of racial prejudice and bias affecting the Black populace in the United States, including in the areas of health, learning and work (Dunklin and Jennings, 2022).

2.2.2 Role of social media in the movement

The Black Lives Matter movement has been significantly boosted by social media in the coordination of protests and the sharing of information (Grimes, 2019). Hashtags such as #BlackLivesMatter on twitter and Facebook has allowed supporters to come together online, to organise themselves and to show support (Guo and Liu, 2022). Such hashtags have been used to set the tone and discussions about the injustices that people of colour face on the digital platforms (Haffner, 2019). With the advancement of the movement, leaders and groups of activists employed social networking to disseminate important information quickly and share information about demonstrations and protests, as well as to put pressure on authorities and decision-makers (Hswen et al., 2021). The virtual nature of the movement also allowed for

addressing the global audience and initiating similar discussions on racism around the world (Jackson, 2019).

As Jiang and Xu (2023) noted, there is the convergence of hashtags such as #BlackLivesMatter and #StopAsianHate on Twitter, which may suggest connections between the campaigns against racism of different ethnicities. Thus, it allowed new forms of decentralized participation as people joined more digital platforms than the traditional top-down organizational structures of past movements (Jones et al., 2022). Social media has been revealed to be quite influential in supporting a youth-led, decentralised movement such as Black Lives Matter to dynamically coordinate and rally supporters at scale.

2.2.3 2.1.3 Participation of women in the movement

Black women have actively engaged in different actions of the Black Lives Matter movement, and some of them have spearheaded the movement. However, their particular involvement and point of view have been a subject of investigation in many cases. Research has revealed that women of colour activists were at the forefront of online activism that sought to address issues such as police violence and racial profiling of Black Americans (Grimes, 2019). Female influencers on Twitter were also key in the agenda setting role by engaging the intersectional feminism in issues to do with gender violence, reproductive and LGBTQ rights, and race-based gender stereotyping (Guo and Liu, 2022).

Jackson (2019) studied the use of hashtags on Twitter and found out that although the overall usage was equal, women focused more on organization, resource, and community support. They also posted more often about their personal experiences of racism and discrimination in everyday life. However, Haffner (2019) observed that women were less frequent than men in the prominent hashtags at the start of the movement, signifying that more work is still required to amplify various voices. Intersectionality acknowledges that factors such as race, gender, class, and sexuality influence how individuals experience oppression and privilege. Therefore, women's activism in the racial justice movement often addresses not only issues of race but also how these issues intersect with their gender and other identities. This approach enriches the understanding of activism by highlighting the diverse and multifaceted experiences of women, emphasizing that their roles are informed by their unique social contexts.

2.2.4 Key issues and campaigns

Perhaps one of the most sustained and ongoing campaigns has been the Black Lives Matter movement, which was founded to address police violence and the racially motivated killings of African Americans by the police (Grimes, 2019). Other related causes that organizations have also funded include the fight against racially motivated hate crimes against Asian Americans (Jones et al., 2022). By engaging both online and on-ground activism, the movement has been able to draw attention to several interrelated issues of systemic racism including police brutality, racially motivated profiling, and economic oppression of the black community.

2.3 Sentiment analysis methods

2.3.1 Lexicon-Based Methods

The use of sentiment lexicons has also been applied to analyze the emotional tone of social movements in the social media, particularly in Twitter. Klein and his team (Klein et al., 2022) used the NRC Emotion Lexicon to analyze the counter-framing approaches of the supporters of the Black Lives Matter movement (Figure 1). They were able to detect different levels of emotional sentiments such as anger, anticipation and trust that is expressed through digital counterpublic communication. However, there are issues with static lexicons, such as the inability to reflect the nuances and shifts in meaning over time (Lampinen, 2020). The meanings attached to slang, misspellings, and new acronyms frequently used in social media may not be captured in general purpose dictionaries (Li et al., 2021).

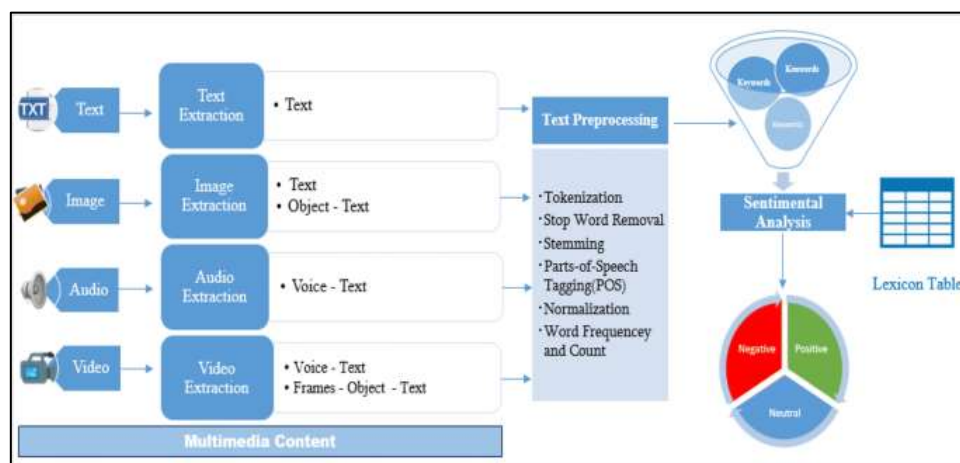


Figure 2.1: A lexicon-based approach for sentiment analysis of multimodal content in tweets Source: (Nartey, 2022)

Many lexicon-based approaches face limitations in analyzing informal social media texts. They often struggle with degree modifiers, which express intensity, as well as negation and

intensification, which can change the meaning of a sentiment. This oversight can lead to inaccurate interpretations of the sentiments expressed in these contexts (Nartey, 2022). For instance, the sarcastic or metaphorical hashtag that may be used to express both support and criticism of the issues under a movement may be misclassified.

Tanksley (2019) also pointed out that when used to examine the social media posts from the open domain, lexicons do not differentiate between sentiments directed towards different targets or topics. In as much as lexicon methods are interpretable because of the sentiment labels used, they are also rigid, not very accurate and limited in flexibility and robustness particularly when analyzing dynamic online conversations. Open platforms can help with the fast annotation of modern colloquialisms and dynamic employments (Lampinen, 2020). However, the process is highly reliant on human input and time, which creates issues with scalability. Some of the promising approaches are the integration of static lexicons with machine learning techniques, which enable lexicon features to be integrated with other dynamic and predictive models during sentiment classification.

2.3.2 Machine Learning Methods

Supervised machine learning has been applied in the analysis of sentiment of social movements on micro blogging sites such as twitter (Figure 2). Some works have used manually labelled tweets to train classifiers to automatically classify tweets into positive and negative sentiments (Nartey, 2022). For instance, Nartey (2022) used supervised learning approaches to analyze framing processes used in counter speech to Black Lives Matter movement on social media. Tong et al. (2022) also used Latent Dirichlet Allocation, an unsupervised topic modelling method to identify new topics that are being discussed within movement related hashtags. However, the efficiency of the machine learning algorithms is greatly influenced by the amount of the annotated data that is available for training (Tanksley, 2019).

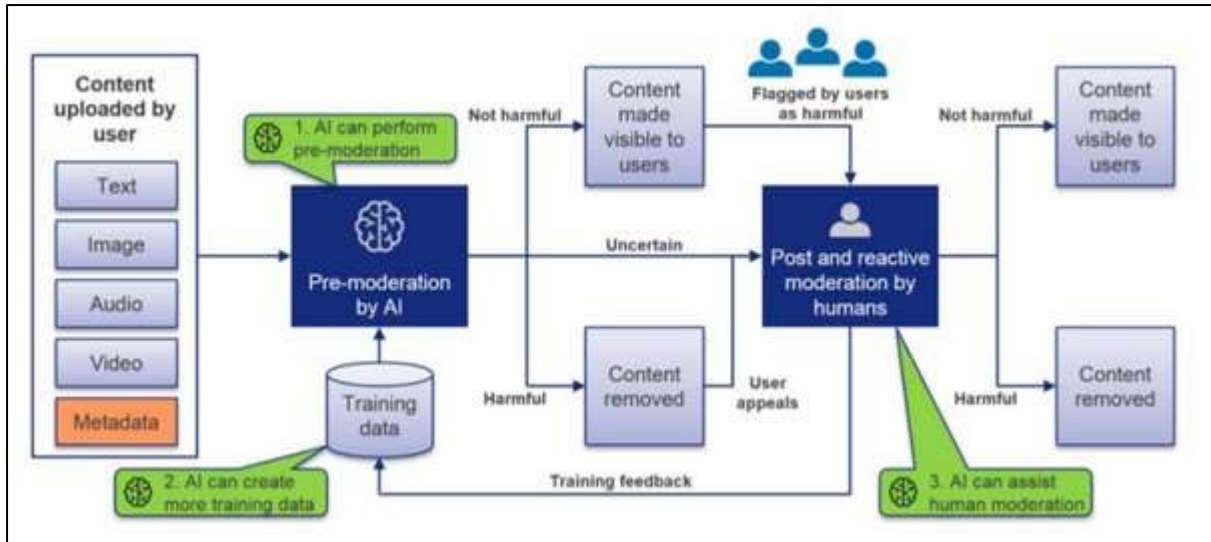


Figure 2.2: How Twitter use machine learning Source: (Li et al., 2021)

Collecting big annotated datasets is time-consuming and involves significant manual labor, especially when focusing on sociopolitical topics in a specific domain (Li et al., 2021). Some challenges arise from the absence of high-quality annotations, and the inconsistency in their availability. Furthermore, most of the classical models fail to capture contextual sentiment in informal, ambiguous social media texts that may contain a range of positions, including nuanced or sarcastic ones (Li et al., 2021). It also includes the feature engineering for the particular task, which might not work well for other related tasks or may not be well scalable to the changes in distribution of the data over time (Toraman et al., 2022). For this reason, the application of the lexicon-based approaches combined with the predictive models can be seen as further perspective. This is because lexicon features may be utilized together with machine learning classifiers that can dynamically update representations from the data (Lampinen, 2020). It can also assist to eliminate several of the drawbacks of individual procedures and improve their stability. However, further research is required for an evaluation of effectiveness of approaches to DA and CD when employing pre-trained models for analyzing the new trends in discussions of social topics in the Internet.

2.3.3 Deep Learning Methods

It is also worthy of note that deep learning techniques have been known to yield high performance in the sentiment classification of texts in social media due to their capability to learn from various levels of abstraction. This is because of the fact that LSTM and GRU bring about recurrent neural networks which have been used to model sequential emotion expressions within movement tweets since they are effective (Li et al., 2021; Schuetze, 2023). For example,

Li et al. (2021) employed RNNs to consider sentiment dynamics in tweets on feminism activism. Other techniques that have been employed are the convolutional neural networks to learn the local contextual patterns as well as using trainable n-gram filters (Schuetze 2023).

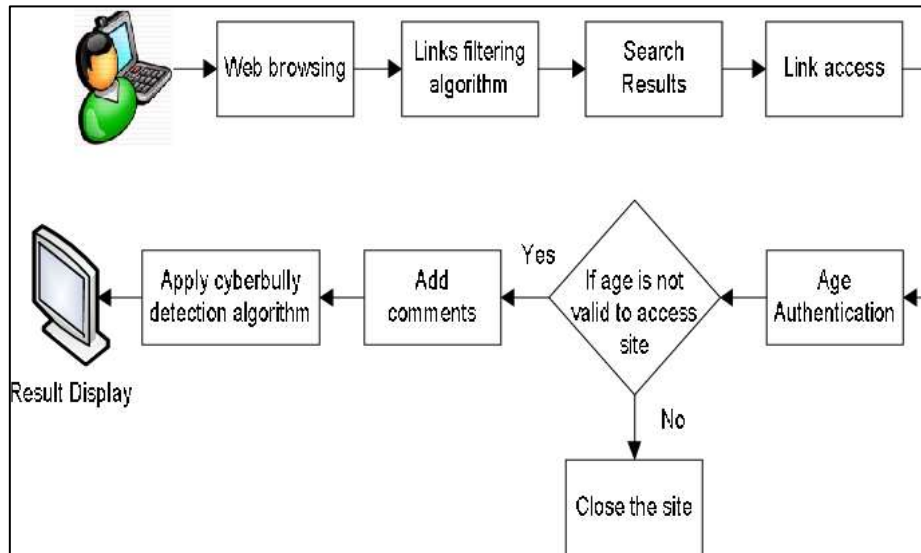


Figure 2.3: Deep Learning Methods tweets Source: (Lampinen, 2020)

The transformer model and its recent improvement in the form of BERT has been identified to be more effective than traditional methods in the recent past. For instance, Leach and Teixeira (2022) applied BERT in the context of the Black Lives Matter movement to estimate the public sentiment from tweets. However, deep learning methods are not without their drawbacks, this is especially due to the fact that they need big annotated sets in order to function and used in the case of specific social and political topics the data set is even harder to come by (Lampinen, 2020; Tanksley, 2019). In addition, deep models can be quite useful when learning highly expressive representations but at the same time, they tend to be black boxes with low interpretability. The integration of deep learning and lexicon-based features together seems to be viable for improved explainability, at worst not causing a performance loss (Lampinen, 2020). However, there are still limitations to reach optimum results and to solve scalability problems and to further improve the sentiment analysis from pre-trained models to identify the sentiment dynamics in the context of ongoing discussions in social media regarding a large number of topics.

2.3.4 Comparison of Different Methods

The performance of the sentiment analysis techniques differs in the extent to which they correctly identify the emotion in the social media texts regarding social movements. The lexicon-based approaches are easily interpretable but have a limited scope and cannot handle

context information (Li et al., 2021; Nartey, 2022). Supervised machine learning utilizes large amounts of unlabelled data, but the accuracy of the model depends on the amount and quality of labelled data (Tanksley, 2019; Tong et al., 2022). Deep learning produces the best results as it learns representations directly but it needs large amounts of data which are hard to obtain (Lampinen, 2020; Leach and Teixeira, 2022).

Combining both lexicon features and predictive models has been reported to help overcome the drawbacks of individual approaches (Lampinen, 2020; Nartey, 2022). For example, integrating semantic information from sentiment lexicons into neural architectures has helped in improving interpretability but at the same time has not negatively impacted the accuracy. The choice of the overall model depends on the data, the complexity of the task, whether interpretability or performance is more important, or both, depending on the data available (Toraman et al., 2022). One of the evaluation dimensions is the capacity to process challenges that are specific to the nature of social media content such as contextual polarity, vagueness, and changes in language/slang usage over time (Li et al., 2021; Klein et al., 2022). Even though deep learning is good at capturing context, the problem of domain adaptation is still a research question. Future work can consider utilizing other methods such as data augmentation to improve the model and its ability to generalize to different open-domain text styles and topics related to social movements on the internet.

2.4 Prior studies on Sentiment Analysis of Social Movements

2.4.1 Analysis of #BlackLivesMatter on Twitter

There has been a significant amount of research on the Black Lives Matter discourse on Twitter. Wilkins et al. (2019) undertook a study of the qualitative analysis of tweets using the #BlackLivesMatter hashtag between 2013 and 2017 to understand the rhetorical roles in framing the movement. In another study, Vitali et al. (2021) examined the social representations of racism and police violence expressed in the #BlackLivesMatter posts on Twitter. In a study conducted by Waterhouse (2022), the researcher explored the political aspect of the platform by analyzing the discourse on the platform after the killing of George Floyd. In their study, Van Haperen et al. (2022) analysed the geographic distribution of clusters and virtual networks of supportive and opposing voices in #BlackLivesMatter conversations on Twitter. In general, these studies offer some understanding of how the movement used social media in advocacy and increasing awareness.

2.4.2 Analysis of other social movements on Twitter

Other studies have also examined sentiment and related discussions about other social movements on the Twitter platform. Vitali et al. (2021) examined the presence of feminist issues in the #MeToo movement on the Twitter platform. Tong et al. (2022) examined the topics from posts with #BlackLivesMatter and #StopAsianHate co-hashtags to investigate the intersectionality. Guo and Liu (2022) discussed agenda-setting effects of hashtag activism between two movements #BlackLivesMatter and #StopAsianHate. Syed and Silva (2023) discussed the sustainability of the Women's March movement on Twitter. The following research papers show how social media is employed for networking, organization, and communication in various causes supporting vulnerable groups.

2.4.3 Gender differences in sentiment and participation

Although earlier research has initiated efforts in describing the overall sentiment and conversation about social movements on Twitter, there has been limited investigation into possible variations based on the participant characteristics. For instance, Vitali et al. (2021) found that #MeToo movement tweets mostly covered issues that are related to feminism but did not examine the author characteristics. Jackson (2019) found that women used Twitter more for organisation/support during Black Lives Matter than men but there is no qualitative analysis of stance, framing or priorities based on gender. Subsequent studies might assist in revealing finer differences by employing both quantitative and qualitative approaches to analyze demographic differences in sentiment, participation, and stance.

2.4.4 Challenges and limitations of prior studies

Previous literature on the study of social movements using Twitter has mainly concentrated on the broad categorization of sentiments, topics and participation networks. Few have considered contextual meaning or the presuppositions that tools alone cannot identify (Waterhouse, 2022). Furthermore, there is a lack of literature on the use of interpretive analysis of diverse characteristics of the participants in the movement, such as exploring them based on gender (Vitali et al., 2021; Jackson, 2019).

2.5 Summary

This chapter has presented a critical review of the research papers in the state of the art. First, the chapter has covered the political movement behind the sentiment. The review also covered the role of social media in modern political movements. The review covered work by researchers aimed at analysing sentiment from social media. The review identified several

limitations such as the difficulty of analysing some social media content from several platforms, and privacy issues. The next chapter presents the research methodology. It gives the plan this research will follow and then implement the experiment.

3 RESEARCH METHODOLOGY

3.1 Introduction

The purpose of this chapter is to provide an overview of the research methodology and the approach taken for this research. It also defines the research philosophy and the approach employed in the analysis of the tweets on the #BlackLivesMatter political movement as experienced by women. Some of the activities covered in this chapter include the method used in the collection and analysis of the data besides the formation of the models using machine learning.

3.2 Research Philosophy

The research philosophy that has been adopted in this study is pragmatism. As the strategy aimed on identifying the type of a real-life problem and using the most suitable research approaches to answer the research question, it was more relevant to adopt a pragmatist paradigm (Saunders, 2015). Therefore, the present study adopted a pragmatic mixed-methods design that included both quantitative and qualitative analysis as the goal was to understand the sentiments that people held about the topic in the social media posts. The qualitative aspect of the research involved the data collected. Tweets are usually textual data, which is qualitative in nature. However, machine learning techniques in the Natural Language Processing field such as lemmatization and tokenization transform the data into quantitative data.

Therefore, this research will solve the problem of effectively understanding public sentiment using the practical solution of analysing social media posts rather than the traditional methods of polls and surveys.

3.3 Research Approach

The research approach is used to guide the researcher on how to reason and arrive at conclusions. There are two main approaches researchers can take. These are deductive and inductive (Saunders 2015). This research will use the inductive research approach. Researchers who use this approach arrive at their conclusion after conducting research and collecting and analysing observations. This contrasts with deductive research where

researchers arrive at conclusions by reasoning from a set of true premises. Therefore, this research will reason from a specific case to a general case, in line with the tenets of inductive reasoning.

This project aims to automate the analysis of public opinion after a politically or socially significant event. This research will cover public sentiment on Twitter for the #BlackLivesMatter movement. The lessons learned in this specific case shall be applied in general cases when events cause intense social media discussions.

3.4 Data Collection and Analysis

The approach used to collect data was to sample all the tweets that are publicly available and had been posted over one year and contained the hashtag #BlackLivesMatter. By employing Selenium, a web automation tool was used to perform web scraping and making it possible to collect new English tweets randomly from different users. The data was then cleaned and normalized and then analysed for feature extraction ready for machine learning. Standardization: In the pre-processing of text data, some techniques that were employed include tokenization, stemming, stop word removal and lemmatization. After that, the filtered training dataset were used in building the machine learning models for the sentiment classification. For instance, LSTM RNNs, BERT transformer networks were designed based on the deep learning approaches which are reliable for achieving high reliability in terms of NLP tasks (Shrestha & Mahmood, 2021).

3.5 Research Design

The study design for the current study was qualitative. The qualitative approach was relevant in order to gain an insight on how individuals express their perceptions and feelings towards the topic of discussion on the social media platform (Creswell & Creswell, 2018). Thus, while the study was concerned with the frequencies, the researchers focused on the contextual meanings, other meanings and the assumptions that would not be retrieved through the computational techniques.

3.6 Research Strategy

The research strategy refers to the procedure the researcher will use to collect and analyse their observations. This study will use the experimental research strategy. This strategy is highly favoured by objective researchers. Additionally, using the experimental strategy creates

reproducible research that can be replicated and verified by other scholars. The experimental strategy involves using a set of dependent and independent variables. The researcher will apply changes called treatments to the independent variable, and then observe how the dependent variables will perform.

This research will have two main variables. These are the algorithms being used and the results they obtain. The experiment will use several algorithms, such as Logistic Regression, XGBOOST, BERT, Random Forest, and the Decision tree. The algorithms are the independent variable because they are applied as treatments to the research. They are the changing variable.. On the other hand, the dataset used to train the models remains the same, making it a constant. However, the values of the assessment metrics depend on the algorithm being used. Thus, these are the dependent variables. The metrics will be discussed further in chapter 4.

3.7 Time horizon

Time horizon refers to the time factor for the research. There are two main types of time horizons. Cross-sectional research indicates that the data was collected at a single point in time. On the other hand, a longitudinal study studies subjects or a phenomenon over a continuous period. This research can be categorized as longitudinal. The data in this research were collected over a period of eighteen months during the #BlackLivesMatter hashtag from 2023 to 2024. The longer timeframe can reveal how opinion morphs as a movement or political event progresses.

3.8 Development Methodology

The development methodology is a standard procedure that is used to ensure experiments are reproducible and rigorous. This research will use the CRISP-DM development process. CRISP-DM is an acronym for Cross Industry Standard Process for Data Mining Hotz (2024). It is a standard process in machine learning projects for the reasons mentioned before – reproducibility and rigour. This development methodology has six stages. This section briefly explains each of these stages.

The first stage is business understanding. This is the point in the project where the researchers will assess the project's feasibility and impact and formulate its objectives. Additionally, a project's plan is created at this stage. This stage was achieved in the first chapter of this research which described the background, motivation, and objectives.

The second stage is data understanding. Researchers obtain their data during this stage. Data acquisition ranges from simple activities such as downloading a readymade dataset to combining several datasets and creating features. Activities in this stage include describing data and ensuring the quality of the dataset. Additionally, the project will conduct an exploratory analysis to expose preliminary insights from the data.

The third stage is the data preparation stage. This is the most time-consuming phase of the methodology. It requires the researcher to select data, combine attributes to create features, and reduce data dimensionality Hotz (2024). This phase is also called feature engineering. The next phase is the modelling. This phase is often achieved with few lines of code because of existing libraries. The libraries contain code abstractions that combine many technical operations into a simple function interphase. For instance, training a model requires the code “`model.train()`”, which is much simpler than implementing a training algorithm. Activities in this phase involve selecting an algorithm, confirming or configuring its parameters, and training the model. Testing the model and assessing it using standard machine learning metrics is also done in this stage.

The fifth phase of the CRISP-DM process is the evaluation phase. This phase reflects on the research and determines whether the project achieved its objectives. Researchers also identify limitations, recommendations, and areas that could use improvement. Finally, the machine learning model will undergo final testing and complementary development such as building a web interface. This is the deployment phase. Clients and the researcher or developer negotiate and plan maintenance strategies during this phase. This research will implement the deployment phase by producing a final report in form of a dissertation.

3.9 Ethical Considerations

Data were collected and handled according to strict privacy and ethical guidelines. Twitter’s privacy was not violated and only public anonymous data that could be used for research was retrieved. All data were kept and analysed safely and were used only for this academic work. Personal information of the users and other details were not disclosed in any manner to avoid compromising the users’ privacy. Bias and fairness were also discussed during the data preprocessing, model construction, and assessment of the model based on the test results with the representative samples, and checking for the potential bias that required improvement during the process (Jobin et al., 2019). The results were presented diligently and were utilized in a responsible manner in order to comprehend the views for the purpose of research rather

than for ill purposes. The principles of research ethical standards of informed consent, data management and social impact were observed.

3.10 Chapter Summary

This chapter has presented the research methodology. The chapter began with a description of the philosophical foundations of this research. These include the research philosophy and approach. It also gave plans for the practical implementation of the research by presenting the research choice, approach, and development methodology. Finally, the chapter described the ethical considerations of this study. The next chapter describes how this research was implemented.

4 IMPLEMENTATION

4.1 Introduction

This chapter aims to implement the plan set out in the previous chapter. The chapter shall begin with a brief discussion of the tools used to conduct the experiment. Then it will describe the data acquisition process and some attributes of the data. This will be proceeded by an exploratory analysis to capture preliminary insights. There will then be a brief feature engineering phase before modelling begins. Finally, the chapter will briefly describe the evaluation metrics used to assess the models.

4.2 Tools used

The following tools were used to conduct this experiment. The tools mentioned in this section are mostly software tools. This research does not mention hardware because the experiment was performed on a standard laptop. The following list shows all the software tools utilized.

1. Python – Python is an interpreted scripting language that is highly favoured by data scientists for its expressiveness. The programming language has a large data science community that builds and maintains software packages for data science.
2. Pandas – Pandas is a data wrangling library used for data manipulation. It can perform tasks such as data descriptions, data cleaning, augmentation, and creation of features.
3. Numpy – Numpy is a software library used to perform efficient numerical calculations, especially on Pandas datasets.
4. Matplotlib – This Python library is used by data scientists to create compelling visualizations that aid in data or observation descriptions.
5. Sci-kit Learn – This Python library contains numerous utilities for sampling, modelling, and assessing a wide range of machine learning models.
6. Selenium – This is a python library that is used for scraping data from websites.

4.3 Data Collection and Understanding

The data acquisition process began with scraping the Twitter website for tweets of the #BlackLivesMatter movement. This activity used the Selenium Web Driver to access the target.

4.3.1 Data Collection Procedure

Navigating the Twitter Login Page

First, the web driver is directed to the Twitter Login page where it provides the name and password for an existing account. One is required by the website to log in before accessing any of the information. The code showing this process is featured in Appendix A.

Figure 4.1 Twitter log in page.

Requesting Tweets

Next, selenium is used to search for tweets related to the #BlackLivesMatter movement. The code showing this process is featured in Appendix B.

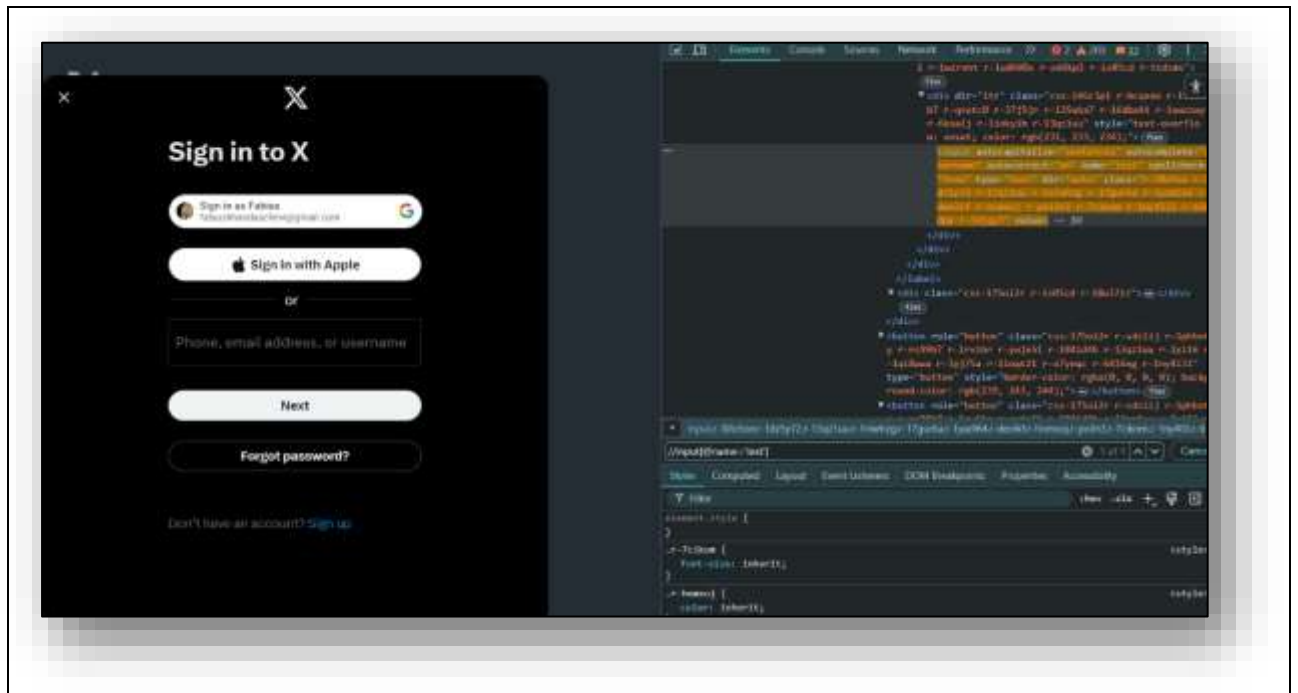


Figure 4.2 tweets requested directly from twitter using selenium.

Extracting Data

The next step was collecting the tweets. Selenium analysed each tweet and extracted the tweet's text along with several metadata. These include the author's username and date information about the tweet. The extracted tweets were stored in a CSV file. Pandas shall be used to manipulate the data as it undergoes preparation. The code showing this activity is featured in Appendix C.

4.3.2 Collected Dataset Information

This section aims to analyse the dataset collected in the previous section. This analysis include the exploratory data analysis for collecting preliminary insights.

Dataset Attributes

The dataset comprises the following key attributes:

User Tags	TimeStamps	Tweets	Replies	Retweets	Likes
Angela Gardner\n@AngieG4bility\n\n12h	2024-06-29T19:44:38.000Z	Investing in our children is the best thing. A...	1	3	16
Black Lives Matter\n@Bklivesmatter\n\n12h	2024-06-29T19:29:52.000Z	Black Boy JOY Alert!\n\nWe're proud to support...	3	29	195
Gabrielle A. Perry, MPH\n@GeauxGabrielle\n\n1d	2024-06-28T14:52:07.000Z	The homeless will be rounded up by police and ...	1.9K	30K	91K
Bryan H., Esq.\n@X_BryanH\n\nJun 28	2024-06-28T14:55:27.000Z	The Supreme Court has\n\nOverturned Roe v. W...	8.5K	40K	136K
Black Lives Matter\n@Bklivesmatter\n\nJun 27	2024-06-27T02:00:23.000Z	CNN's exclusion of Black-owned media from the ...	8	39	69

Figure 4.3 Collected data set attributes.

1. User Tags: This column contains the user's name and Twitter handle. The information is concatenated with line breaks.
2. Timestamps: This column contains the time when the tweet was posted.
3. Tweets: This column contains the tweet's text
4. Replies: This column shows the number of replies a tweet received.
5. Retweets: This column indicates the number of retweets for each tweet.
6. Likes: This column indicates the number of likes for each tweet.

4.4 Data Pre-processing

Data pre-processing is crucial for ensuring the dataset is clean and suitable for analysis. The following steps are undertaken:

4.4.1 Data Cleaning

The data processing process comprised three steps. The first was removal of duplicates to maintain unique entries. Next, the experiment handled the missing values through imputation. However, some rows were removed from the dataset. The final step is parsing dates from timestamps to a standard date time format.

Text Pre-processing

1. Cleaning Tweet Text: URLs, mentions, hashtags, and special characters were removed, and the text was converted to lowercase.
2. Tokenization: The tweet text was split into individual tokens.
3. Lemmatization: Words in the tweets were reduced to their base form.
4. Common stop words that are not useful were removed for analysis.

The code showing these data cleaning activities is shown in Appendix D.

4.5 Feature Engineering

4.5.1 Feature extraction

Feature engineering started with extracting features from the timestamp, such as the day of the week, hour of the day, etc.

Sentiment Analysis

The sentiment analysis process involved assigning sentiment labels to the tweets in the dataset. Using a sentiment analysis tool, each tweet was evaluated and given a sentiment score. These scores were then categorized into Positive, Negative, and Neutral sentiment labels.

Sentiment scores were assigned to tweets based on their content. Tweets with a sentiment score greater than zero were labelled as "Positive.", tweets with a zero-sentiment score were labelled as "Neutral.", while tweets with a sentiment score equal to zero were labelled as "Negative."

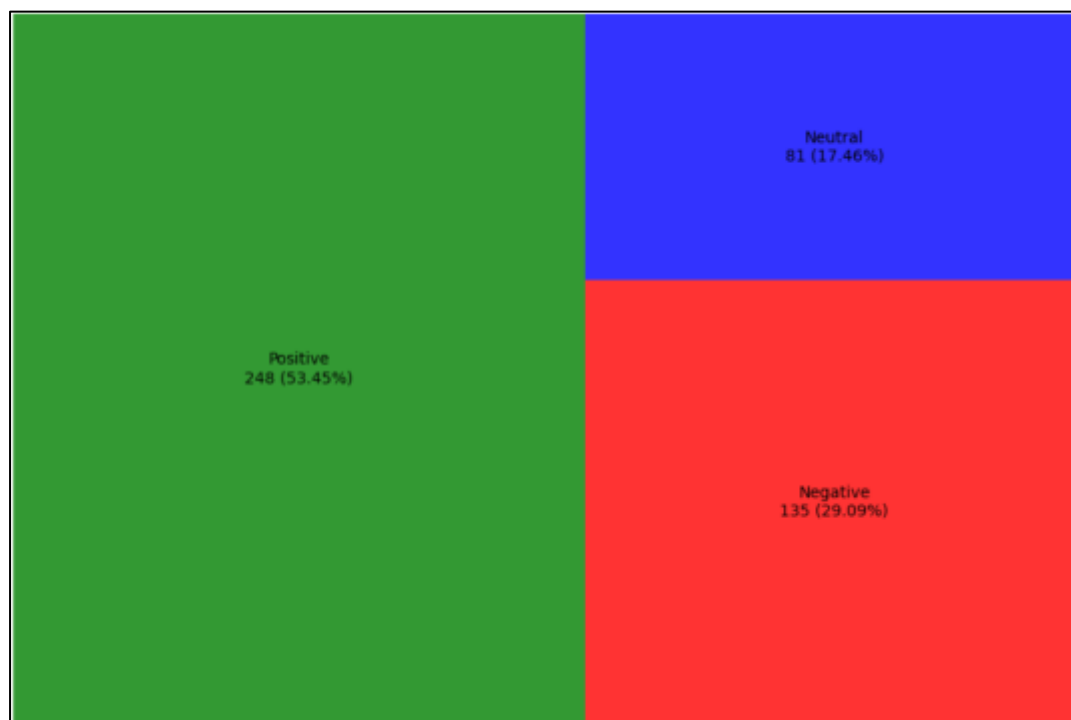


Figure 4.4: Distribution of sentiment labels

This distribution indicates that most of the discussions in the dataset were positive, reflecting a supportive or hopeful outlook among the users. However, many negative tweets highlight ongoing concerns or criticisms within the movement. The smaller percentage of neutral tweets suggests that most users had a clear positive or negative stance on the issues discussed.

The code that performed this analysis is shown in Appendix E.

4.5.2 Data Transformation

Vectorization: Cleaned text was converted into numerical vectors using TF-IDF or word embeddings. Numerical features like replies, retweets, and likes were normalized in a process called normalisation.

This is the new dataset as a result of pre-processing.

	UserTags	TimeStamps	Tweets	Replies	Retweets	Likes	Cleaned_Tweets	Sentiment
0	Angela Gardner/n@AngleG4ability/n/n12h	2024-06-29 19:44:38+00:00	Investing in our children is the best thing. A...	1.0	3.0	16.0	investing in our children is the best thing an...	1.000000
1	Black Lives Matter/n@Bklivesmatter/n/n12h	2024-06-29 19:29:52+00:00	Black Boy JOY Alert/n/nWe're proud to support...	3.0	29.0	195.0	black boy joy alert/n/nwere proud to support t...	0.522222
2	Gabrielle A. Perry, MPH/n@GeauzGabrielle/n/n1...	2024-06-28 14:52:07+00:00	The homeless will be rounded up by police and ...	1900.0	30000.0	91000.0	the homeless will be rounded up by police and ...	0.000000
3	Bryan H., Esq./n@X_BryanH/n/nJun 28	2024-06-28 14:55:27+00:00	The Supreme Court has/n/nOverturned Roe v. W...	8500.0	40000.0	136000.0	the supreme court has/n/noverturned roe v made...	0.350000
4	Black Lives Matter/n@Bklivesmatter/n/nJun 27	2024-06-27 02:00:23+00:00	CNN's exclusion of Black-owned media from the ...	8.0	39.0	69.0	cnn's exclusion of blackowned media from the pr...	0.100000

Figure 4.5New data set after pre-processing.

4.5.3 Data Analysis

Data analysis involves exploring and understanding the dataset to extract meaningful insights and identify patterns. Here, the focus was on the data collected from tweets about the #BlackLivesMatter political movement among women. This analysis will help us understand the distribution and characteristics of the data, providing a foundation for further modelling and evaluation.

Basic Statistics

Basic statistics such as mean, median, and standard deviation were calculated for numerical columns like replies, retweets, and likes. This provides a quantitative summary of the dataset.

The following analysis shows various visualisations for word lengths in tweets. This analysis might provide insights into the nature of the tweets' authors during the period

a) Histogram of Word Lengths in Tweets

A histogram was created to visualize the tweets' overall distribution of word lengths.

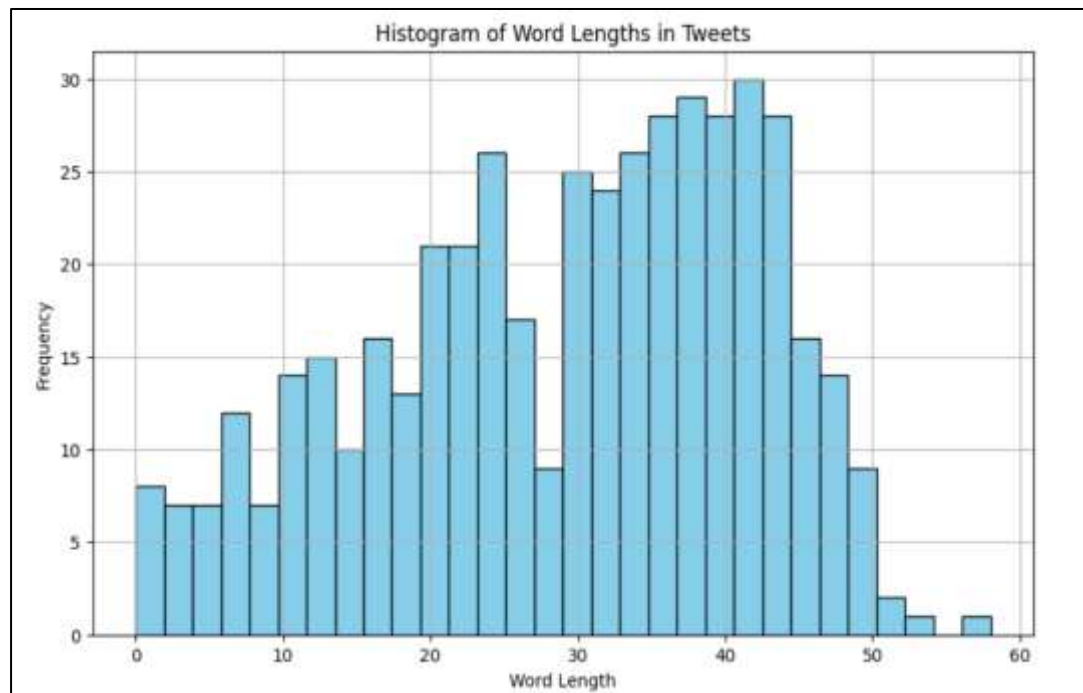


Figure 4.8 Histogram of Tweet's overall distribution.

The analysis shows that most tweets have between 35 and 40 words, showing that most tweets are short and most likely about a single concern or topic.

b) Box Plot of Word Lengths by Sentiment

The following box plot compares the lengths and their variances of tweets with respect to their sentiments.

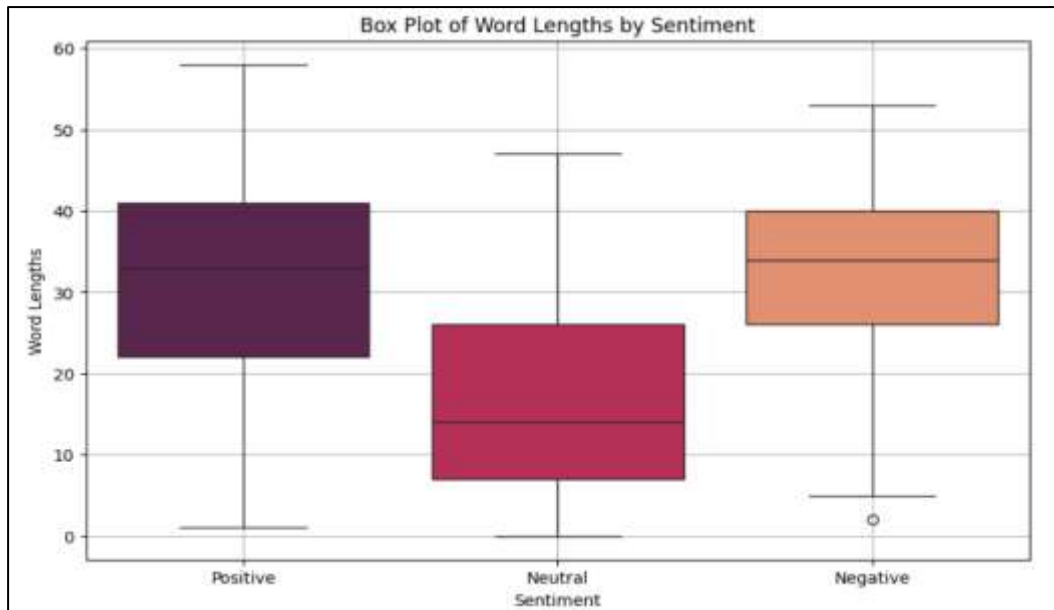


Figure 4.9 Box plot comparing word length of tweets across different sentiment categories.

The analysis shows that positive tweets tend to be longer on average than neutral and negative tweets. These results could mean that positive tweets are mainly responding and giving facts to accusations levelled by negative tweets. The results could also show the influence of misinformation.

Trend Analysis of Daily Tweets

The trend of daily tweets was plotted to examine the activity pattern and frequency of tweets over time.

The line plot shows the number of tweets posted each day. This helps identify any peaks or patterns in tweet activity over time. The visualization indicates significant events or periods with increased activity related to the #BlackLivesMatter movement.

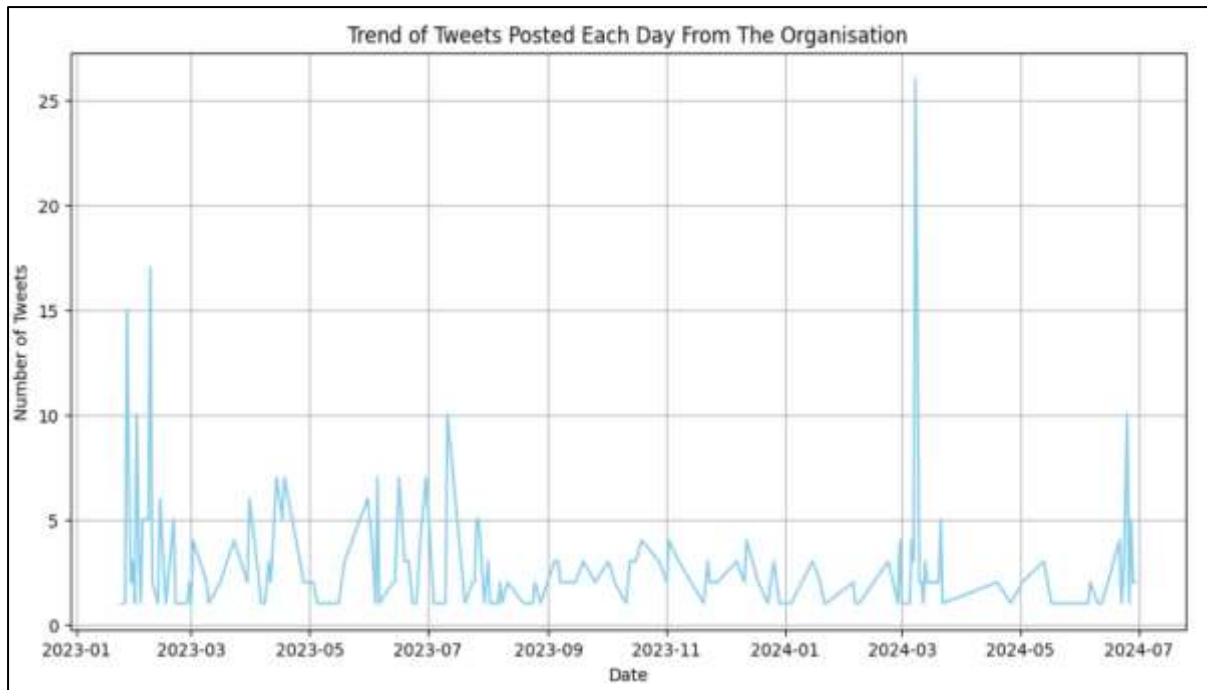


Figure 4.10 Line Plot of tweets posted each day from the organisation.

As seen from the line plot, there are several peaks, with March recording the highest tweets posted from the movement. These suggest days of heightened social media engagement. Valleys (where the line drops to nearly zero) represent days with minimal or no tweeting.

4.5.5 Twitter Metrics Analysis

Correlation Matrix of Likes, Retweets, and Replies

The heatmap visualization displays the correlation matrix between the three key metrics: Likes, Retweets, and Replies. The heatmap provided is helpful in identifying which metrics are more likely to influence each other, providing insights into user engagement behaviours on Twitter. The matrix is as shown below:

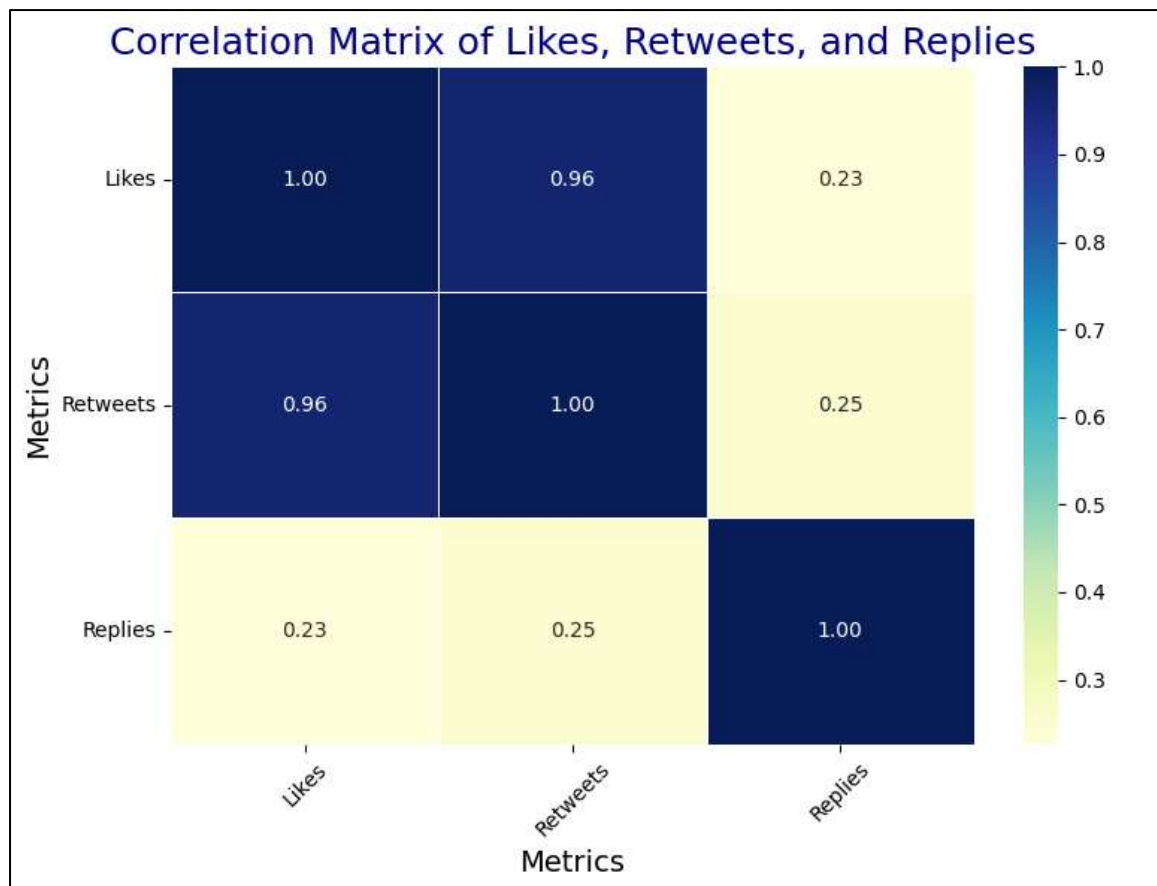


Figure 4.11 Correlation Matrix of likes, Retweets, and Replies.

The matrix shows a very strong positive correlation (0.96) between Likes and Retweets. This suggests that tweets with more likes tend to have more retweets. The correlation between Likes and Replies is relatively low (0.23), indicating that the number of likes a tweet receives is not strongly related to the number of replies. The correlation between Retweets and Replies is also low (0.25), suggesting that tweets that are retweeted frequently do not necessarily receive many replies.

Distribution of Tweets, Likes, Retweets, and Replies by Hour

The visualizations below provide insights into the hourly distribution of tweets, likes, retweets, and replies.

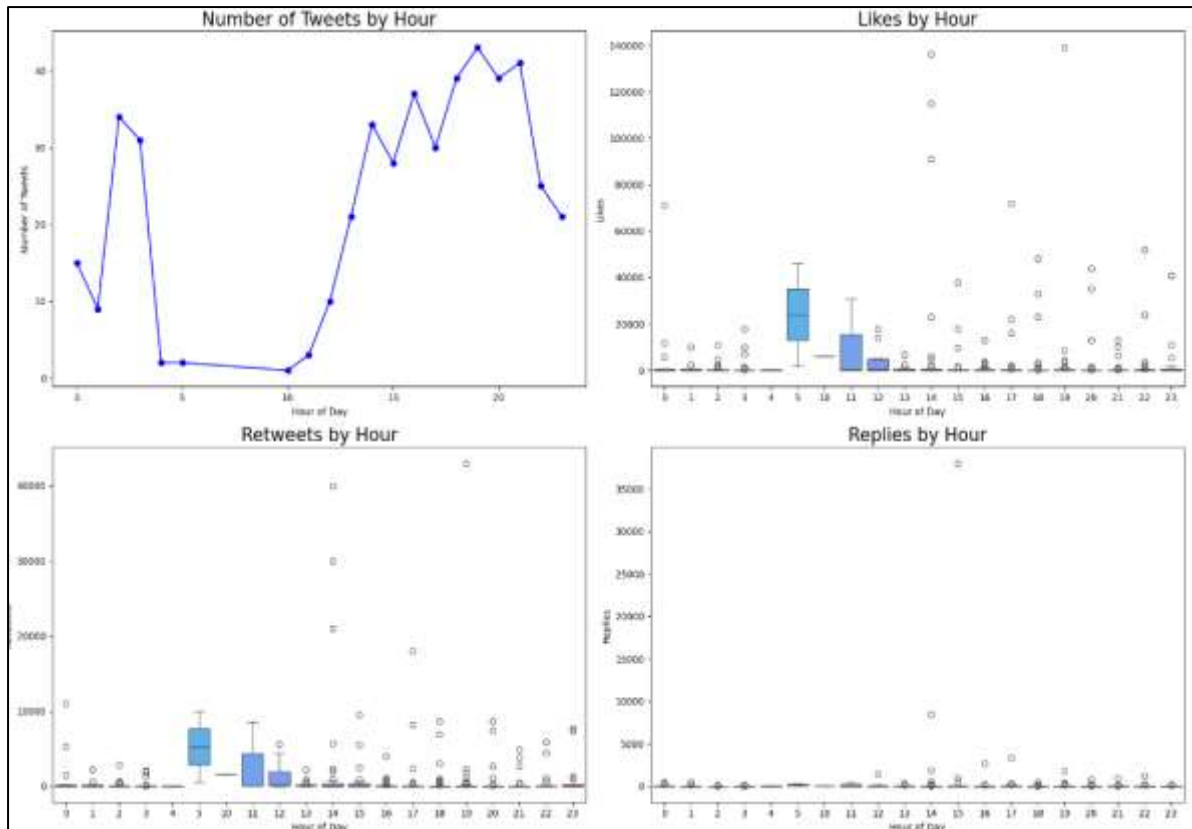


Figure 4.12 Visualizations an hourly distribution of tweets ,likes, retweets and replies.

The following insights are gathered from the above graphs:

1. Number of Tweets by Hour

The line plot indicates that the number of tweets varies throughout the day. There is a noticeable dip in the early morning hours (around 5 AM) and a peak in the evening (around 8 PM to 10 PM).

2. Likes by Hour

The box plot indicates the distribution of likes received by tweets posted at different hours. Tweets posted around 3 AM and 11 AM tend to receive more likes, as noted in the higher interquartile range and median.

3. Retweets by Hour

The box plot for retweets mirrors the pattern observed for likes, with notable peaks around the same hours (3 AM and 11 AM). This aligns with the earlier observed high correlation between likes and retweets.

4. Replies by Hour

The distribution of replies is more dispersed and does not show as clear a pattern as likes and retweets. However, there are two outliers in the afternoon hours which could indicate the time most users are active on the site

These visualizations provide a detailed overview of user activity related to the hashtag and engagement patterns throughout the day.

Analysis of Engagement Metrics

This can be illustrated by the histogram plots shown below:

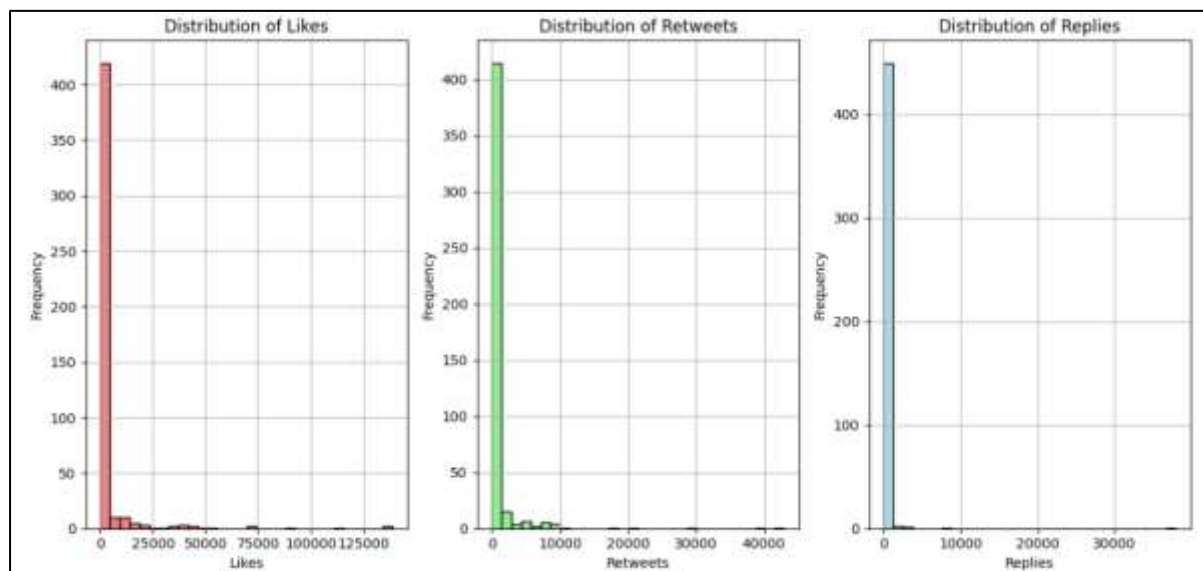


Figure 4.13 Histogram illustrating analysis of engagement metrics.

1. Distribution of Likes

The plot shows that a vast majority of tweets have low numbers of tweets. This is typical of social media posts where few popular accounts get most engagements while most people get a few likes from mostly immediate friends and family.

2. Distribution of Retweets

Most tweets have a low retweet count, indicating that viral tweets (those with high retweet numbers) are relatively rare. This distribution highlights the disparity in engagement, with a few tweets achieving significant reach through retweets.

3. Distribution of Replies

The third subplot displays the distribution of replies. This distribution also follows a pattern where the majority of tweets receive a low number of replies. As with likes and retweets,

the frequency of tweets decreases sharply as the number of replies increases. This pattern underscores that high engagement in terms of replies is uncommon, with most tweets attracting minimal direct responses.

This trend, shown by the engagement metrics from the histograms, is very common in social media data, where viral content represents a small fraction of the total content. These insights are crucial for understanding user behaviour and the nature of content that resonates widely within the #BlackLivesMatter movement among women.

Top 10 Hashtags in Black Lives Matter Movement Tweets

The bar chart displaying the top 10 hashtags reveals the most frequently used tags in tweets related to the Black Lives Matter movement. The most common hashtags are #SOTU, #NAACPxSOTU, and #sotu2023, which indicate key events or topics of interest within the movement.

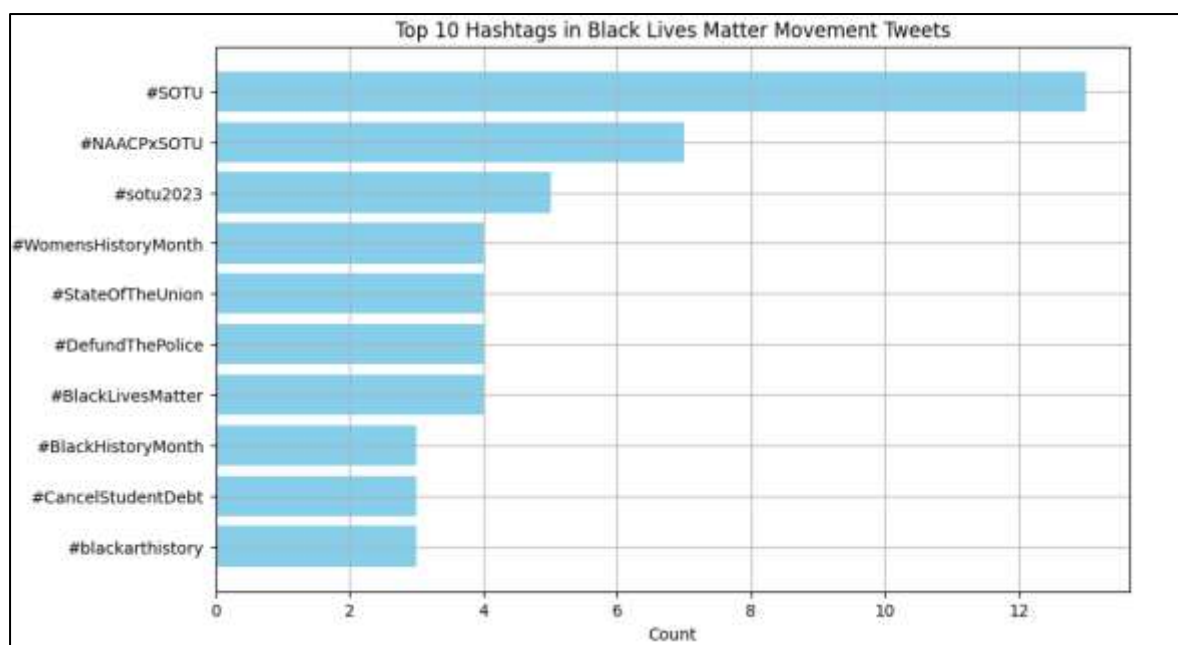


Figure 4.14 Barb chart displaying top 10 hashtags.

Top 10 Mentions in #BlackLivesMatter Tweets

The pie chart below shows the top 10 mentions and provides insight into tweets' most frequently mentioned users. Accounts like @POTUS, @Cicley_Gay, and @WhiteHouse are among the top mentioned, suggesting the involvement and recognition of prominent political figures and organizations in the discourse surrounding the Black Lives Matter movement.

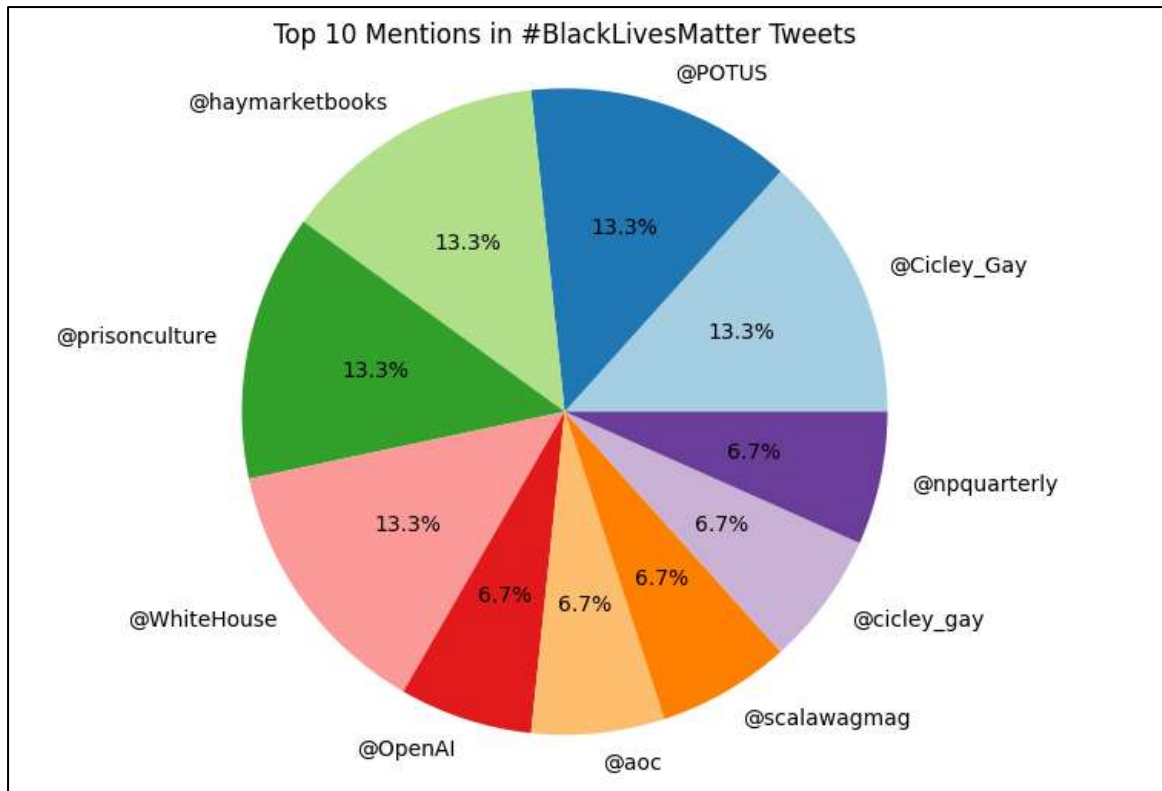


Figure 4.15 Pie chart displaying top 10 mentions in the tweets

4.6 Modelling

4.6.1 Logistic Regression

The Logistic Regression model is one of the most popular algorithms for binary and multiclass classification problems. The algorithm works by performing a logit transformation, which means the probability of success is divided by the probability of failure (LaValley 2008). The resulting probability measure shall be between 0 and 1. Binary classifiers can set a threshold such as 0.5 to distinguish between classes. It is imperative to test the model for good fit and overfitting.

Data Vectorization

The TF-IDF (Term Frequency-Inverse Document Frequency) vectorizer converted text data into numerical form.

TF-IDF helps in transforming text data into a matrix of TF-IDF features, which reflects the importance of a word in a document relative to a collection of documents (corpus).

Model Training

A Logistic Regression model was trained using the training set. The `max_iter` parameter was set to 1000 to ensure convergence during training. The code showing this activity is shown in Appendix F

4.6.2 XGBoost (*Extreme Gradient Boosting*)

XGBoost is an optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. It implements machine learning algorithms under the Gradient Boosting framework. It has the advantage of handling missing data inherently and providing feature importance scores, which can be helpful for feature selection. Its regularization mechanisms (L1 and L2) help prevent overfitting XGBoost, which was selected due to its high performance in competitions and real-world applications, often outperforming other algorithms. It is also highly flexible and can handle classification and regression tasks efficiently.

Similar to Logistic Regression, the text data was converted into TF-IDF vectors. The XGBoost model was trained using the TF-IDF vectors of the training set. The `eval_metric` parameter was set to 'mlogloss' (Multiclass Logarithmic Loss) to evaluate the performance during training. The code showing the XGBOOST modelling is shown in Appendix G.

4.6.3 BERT (*Bidirectional Encoder Representations from Transformers*)

BERT is a transformer-based model designed to understand the context of a word in search queries. It is particularly well-suited for natural language processing tasks. BERT's main advantages include the availability of pre-trained models that can be fine-tuned for specific tasks, saving computational resources, and its ability to generate contextualized word embeddings, which provide a better understanding of words in context (Devlin 2018). However, BERT is resource-intensive, requiring significant computational power and memory, especially for training. Fine-tuning can also be time-consuming.

1. Tokenization

Text data was tokenized using the BERT tokenizer, which converts the text into a format suitable for BERT (i.e., tokens and attention masks).

2. Data Encoding

The tokenized data was encoded into input IDs and attention masks and then converted into tensors for training.

3. Model Training

The BERT model was fine-tuned using the encoded data.

Training parameters included a specified number of epochs, batch size, and learning rate settings defined in the Training Arguments.

The code snippet showing the BERT Modelling is shown in Appendix H.

4.6.4 *Random Forest*

Random Forest is an ensemble learning method that operates by constructing multiple decision trees during training and outputting the mode of the classes (classification) of the individual trees. It can capture complex interactions between features and provide insights into their importance. Random Forest was selected for its robustness and versatility.

Textual data was converted into numerical form using the TF-IDF vectorizer. The model was configured using to use 100 estimators. The code snippet showing the Random Forest Training is shown in Appendix I

4.6.5 *Decision Tree*

Decision Tree is a supervised learning algorithm used for classification and regression. It works by making minimal assumptions about the nature of the data its modelling, thus making it non-parametric. The algorithm works by dividing data instances into several decisions. This algorithm works best when there are few decisions, which reduces the depth and complexity of the tree. Complex decision trees could suffer from overfitting.

4.6.6 *Data Vectorization*

Text data was converted into numerical form using the TF-IDF vectorizer.

The Decision Tree model was trained using the newly transformed dataset. The model was configured with a random state to ensure reproducibility.

4.7 Evaluation metrics

This section aims to present the evaluation metrics used to assess the model's performance during testing. The metrics used are standard machine learning metrics used to assess how good model predictions are. These metrics are:

1. Accuracy – Accuracy simply measures the ratio of correct predictions to all predictions. It doesn't matter if the predictions are positive or negative.
2. Precision – precision is used to assess the ratio of true positive predictions to the total number of precision. This metric aims to measure how often models are correct when making a positive prediction.

3. Recall – Recall measures the ratio of true positive prediction to all positive instances. It therefore measures the proportion of positive instances that the model will capture.
4. F1 score – This metric measures how balanced a model is. It is the harmonic mean of precision and recall, meaning deviation in one will significantly affect the score.

4.8 Chapter Summary

This chapter has explained how this research implemented its experiment. It followed the plans laid out in the previous chapter. The chapter began with a description of the tools and data. Next, it performed an exploratory data analysis that identified preliminary insights on the data. Next, the experiment performed modelling after feature engineering. Finally, the chapter described the evaluation metrics. The next chapter provides the results of the experiment.

5 RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents findings from the sentiment analysis of tweets related to the #BlackLivesMatter political movement among women. It compares and analyses the performance of different machine learning models—Logistic Regression, XGBoost, Random Forest, Decision Tree, and BERT. The implications of these results and possible future work are also discussed.

5.2 Model Performance

This section shows how the models performed in terms of the evaluation metrics described at the end of chapter 4.

5.2.1 Logistic Regression

The following results show the performance of the logistic regression model.

Metric	Value
Accuracy	52.69%
Precision	0.51
Recall	0.36
F1-Score	0.27

Table 5.1 Performance metrics for the Logistic Regression.

The model Struggles to accurately distinguish between positive and negative tweets about the #BlackLivesMatter movement. While it captures some accurate positive tweets (moderate precision), it misses a significant portion of positive tweets (low recall) and misclassifies some tweets (lower accuracy). There might be a bias toward identifying negative or missing positive tweets altogether (reflected in the low F1 score).

Classification Report Plot Matrix for the Logistic Regression Model

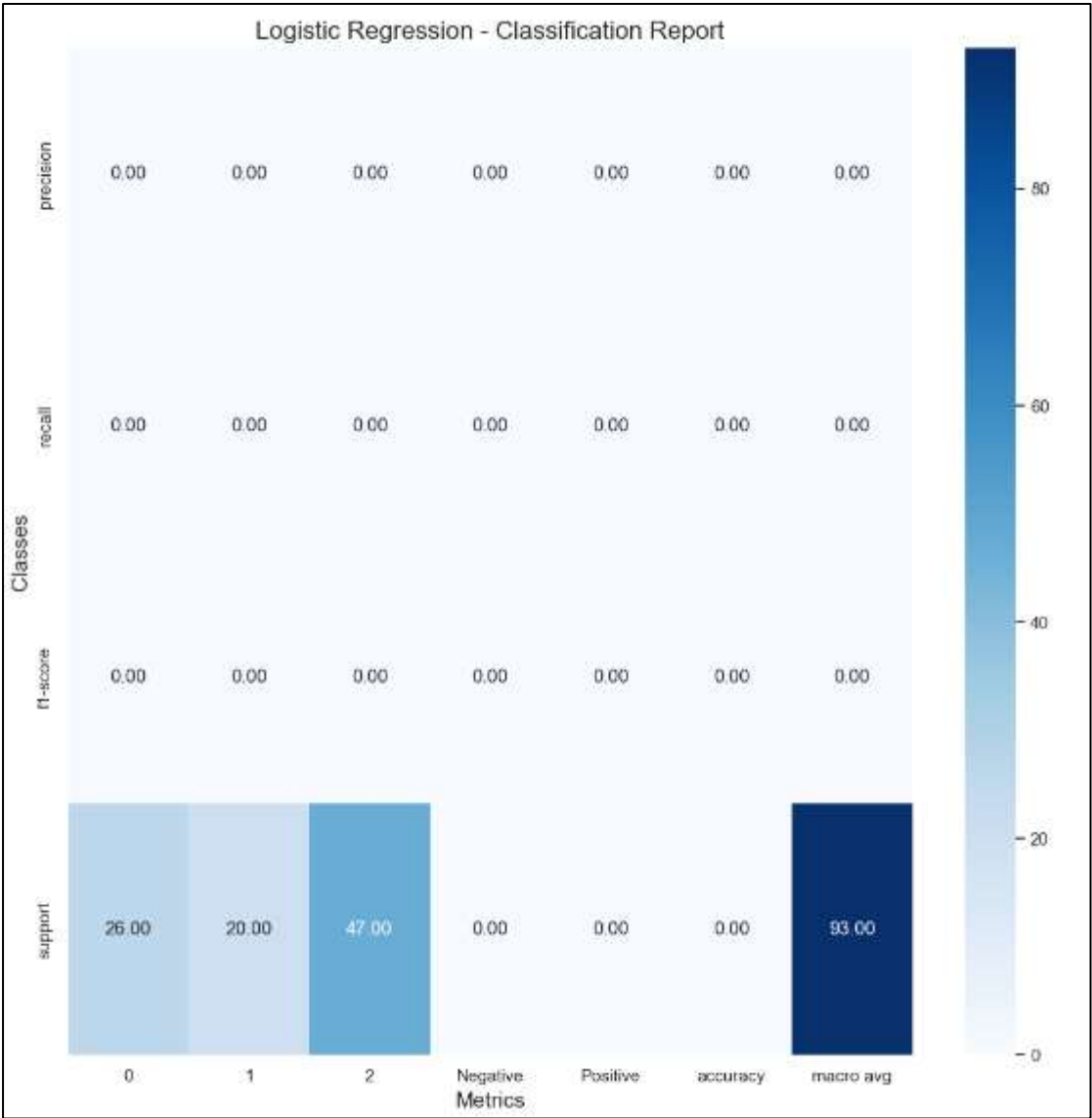


Figure 5.1 Classification Report for the Logistic Regression Model.

5.2.2 XGBoost

The performance metrics for this model are as follows:

Metric	Value
Accuracy	58.06%
Precision	0.59
Recall	0.58

F1-Score	0.57
----------	------

Table 5.2 Performance metrics for this model.

The classification report below visualises the model's ability to accurately predict which class tweets belongs to.

Classification Report Plot Matrix for the XGBoost Model

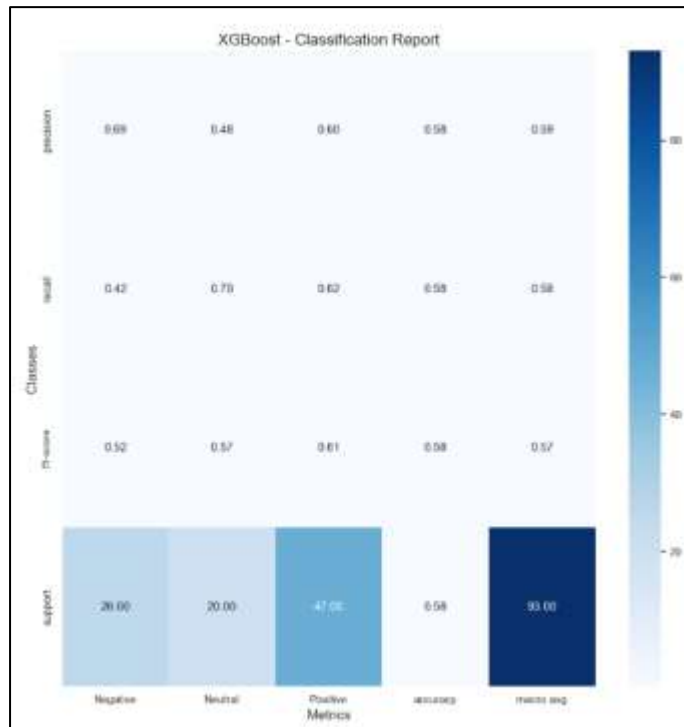


Figure 5.2 Classification Report Plot Matrix for the XGBoost Model.

The XGBoost model performs closely to the Random Forest model analyzed in terms of accuracy (58.06% vs. 59.14%). Compared to the Logistic Regression model, XGBoost is more accurate, precise, recall, and F1-score, indicating a better overall performance in capturing the sentiment of tweets related to the #BlackLivesMatter movement. XGBoost struggled to classify many tweets, suggesting room for improvement. The XGBoost model performs reasonably well in classifying sentiments of tweets related to the #BlackLivesMatter movement among women. Its balanced metrics across accuracy, precision, recall, and F1-score suggest it can reliably identify and differentiate between sentiment classes.

5.2.3 BERT

The performance metrics by the BERT model for the sentiment analysis are as shown below.

Metric	Value
Accuracy	51.61%
Precision	0.50
Recall	0.35
F1-Score	0.26

Table 5.3 Performance metrics by BERT model.

The following graph shows the training and validation loss for the model.

The BERT model exhibited moderate performance. The relatively low precision and recall suggest that while the model can somewhat identify positive sentiment, it definitely struggles with neutral and negative feelings. The low F1 score reflects the model's difficulty balancing precision and recall across different classes.

Training and Validation Loss

The training and validation loss for the BERT model over three epochs are shown below:

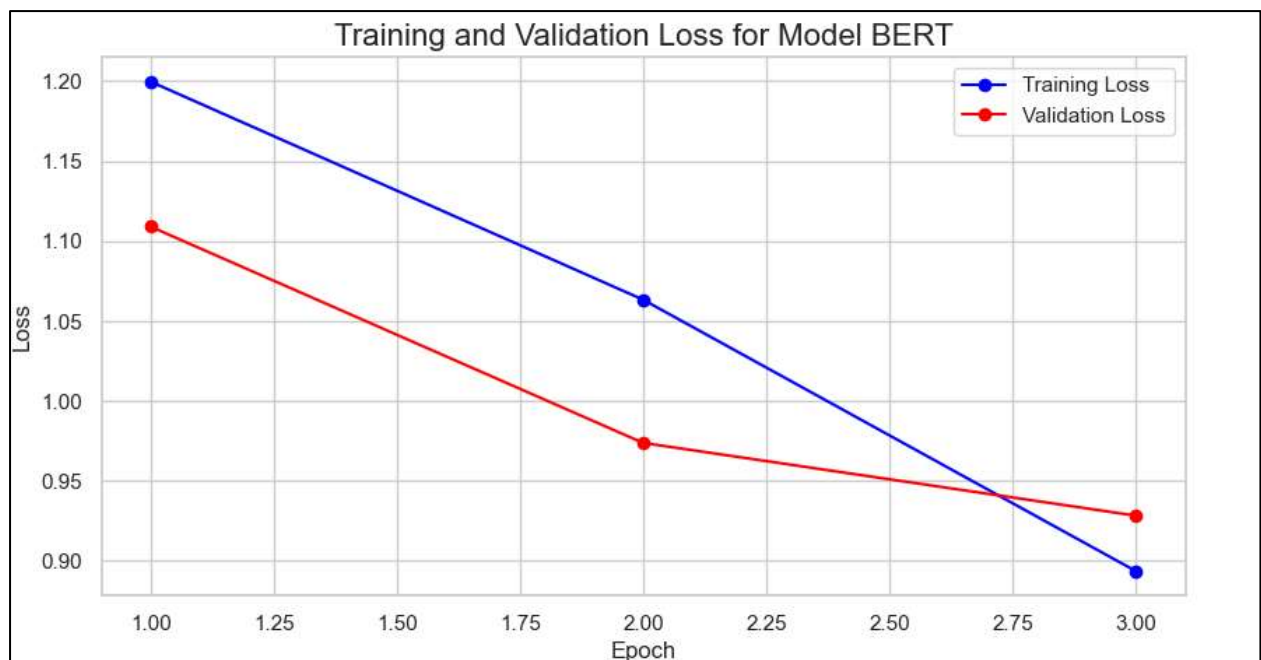


Figure 5.3 line graph representing training and validation loss for model BERT.

Possible reasons for moderate performance

Results of BERT in Training.

The BERT model demonstrates a consistent decrease in both training and validation loss across the epochs. This indicates:

1. **Model Learning:** Effective learning from training data, with the model gradually minimizing error by adjusting internal parameters through backpropagation.
2. **Generalization:** Good generalization of unseen validation data, suggesting that the model does not overfit.
3. **Convergence:** The model converges, approaching optimal parameter values with reduced likelihood of further significant loss improvement.
4. **Performance Stability:** Stable performance, avoiding overfitting, as indicated by the slight difference between training and validation loss.

The BERT model's performance in sentiment analysis of tweets from the #BlackLivesMatter political movement among women shows its capability to learn effectively from the data while maintaining good generalization. Despite not achieving the highest accuracy, its robustness and stability make it a valuable model for this task.

5.2.4 Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees and merges them to obtain a more accurate and stable prediction.

Metric	Value
Accuracy	59.14%
Precision	0.67
Recall	0.51
F1-Score	0.50

Table 5.4: Random Forest representing multiple trees and merges.

Accuracy (59.14%): While it's higher than random guessing (50%), it suggests the model misclassifies nearly 41% of the tweets.

Precision (0.67): This metric measures the proportion of tweets predicted as positive by the model that were positive. A value of 0.67 indicates that for every ten tweets the model classified as positive, roughly seven were truly positive, while three were incorrectly classified.

Recall (0.51): This metric measures the proportion of actual positive tweets the model correctly identified. A value of 0.51 indicates that the model captured about half of the positive tweets in the data, missing nearly 49%.

F1-Score (0.50): This metric is a harmonic mean of precision and recall, providing a balanced view of both. The F1-score of 0.5 aligns with the lower recall, suggesting the model might be biased towards precision.

Classification Report Plot Matrix for the Random Forest Model

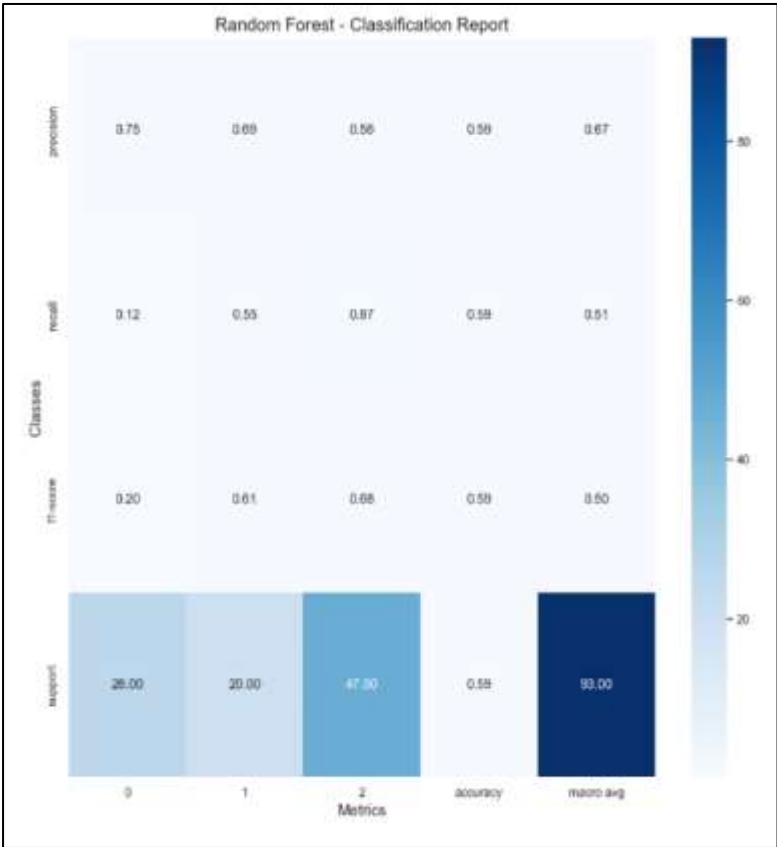


Figure 5.4 Classification Report Plot Matrix for the Random Forest Model.

The Random Forest model's performance metrics reflect moderate effectiveness in sentiment analysis. With an accuracy of 59.14%, the model demonstrates a basic understanding of sentiment patterns in tweets from the #BlackLivesMatter political movement among women.

5.2.5 Decision Tree

Metric	Value
Accuracy	51.61%
Precision	0.52
Recall	0.55
F1-Score	0.52

Table 5.5 Decision tree representing the most significant attribute.

Accuracy (51.61%): This represents the overall proportion of tweets the model classified correctly (positive or negative). While it might seem low at first glance, it's essential to consider the baseline performance (50%) for a random guess. However, a score above 50% still suggests room for improvement.

Precision (0.52): This metric measures the proportion of tweets predicted as positive by the model that were positive. A value of 0.52 indicates that for every 100 tweets the model classified as positive, roughly 52 were truly positive, while 48 were incorrectly classified.

Recall (0.55): This metric measures the proportion of actual positive tweets the model correctly identified. A value of 0.55 indicates that the model captured roughly 55% of the positive tweets in the data while missing 45%.

F1-Score (0.52): This metric is a harmonic mean of precision and recall, providing a balanced view of both. The F1-score of 0.52 aligns with the precision value, suggesting that the model might be biased towards classifying tweets as positive.

Classification Report Plot Matrix for the Decision Tree Model

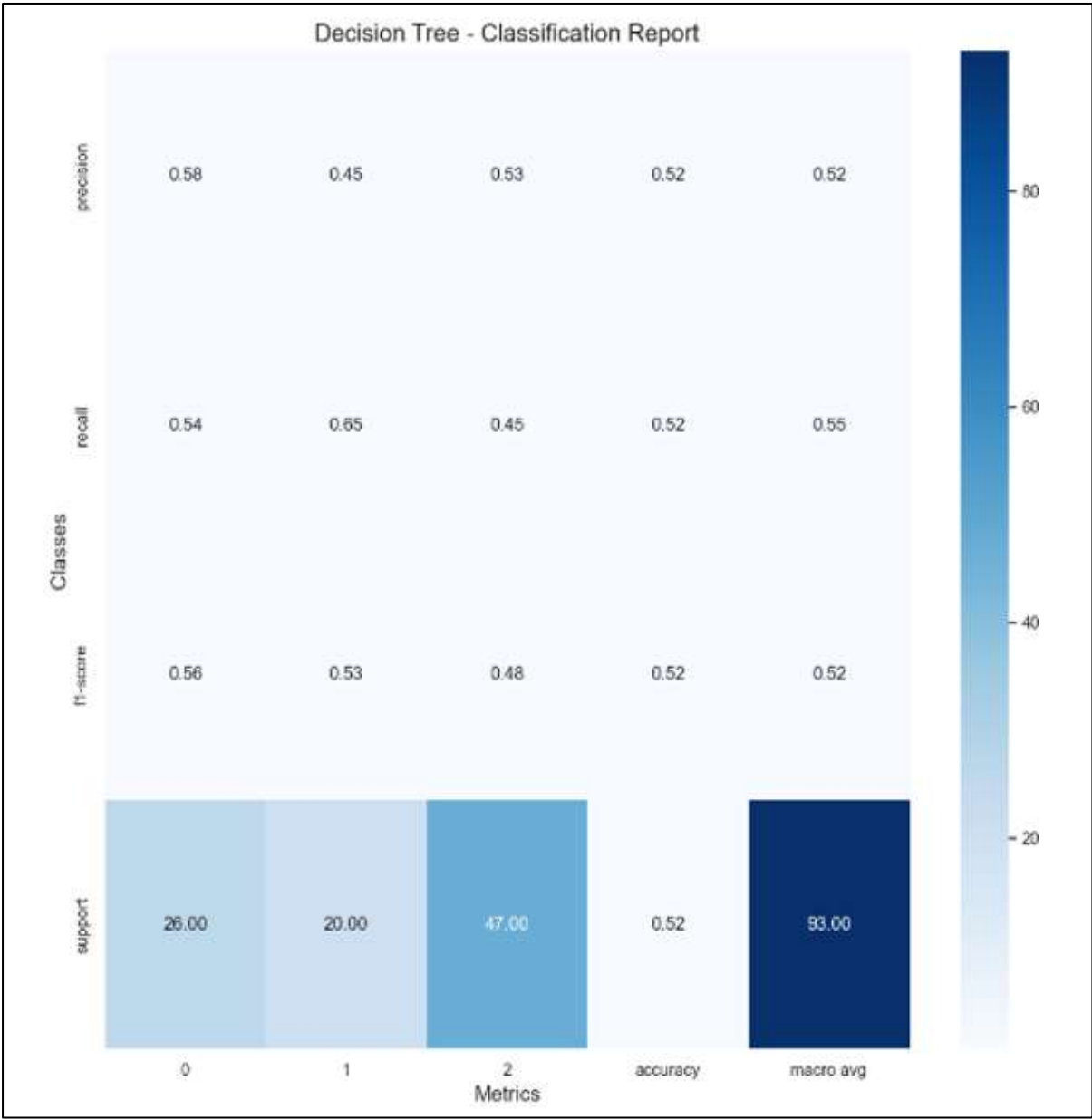


Figure 5.5 Classification Report Plot Matrix for the Decision Tree Model.

The Decision Tree model demonstrates moderate performance in classifying sentiments of tweets related to the #BlackLivesMatter movement among women. While it shows consistency between precision and recall, its overall accuracy suggests substantial room for improvement. This performance level indicates that the model may be unreliable for nuanced sentiment analysis tasks without further refinement.

5.3 ROC Curve Analysis for Sentiment Analysis Models

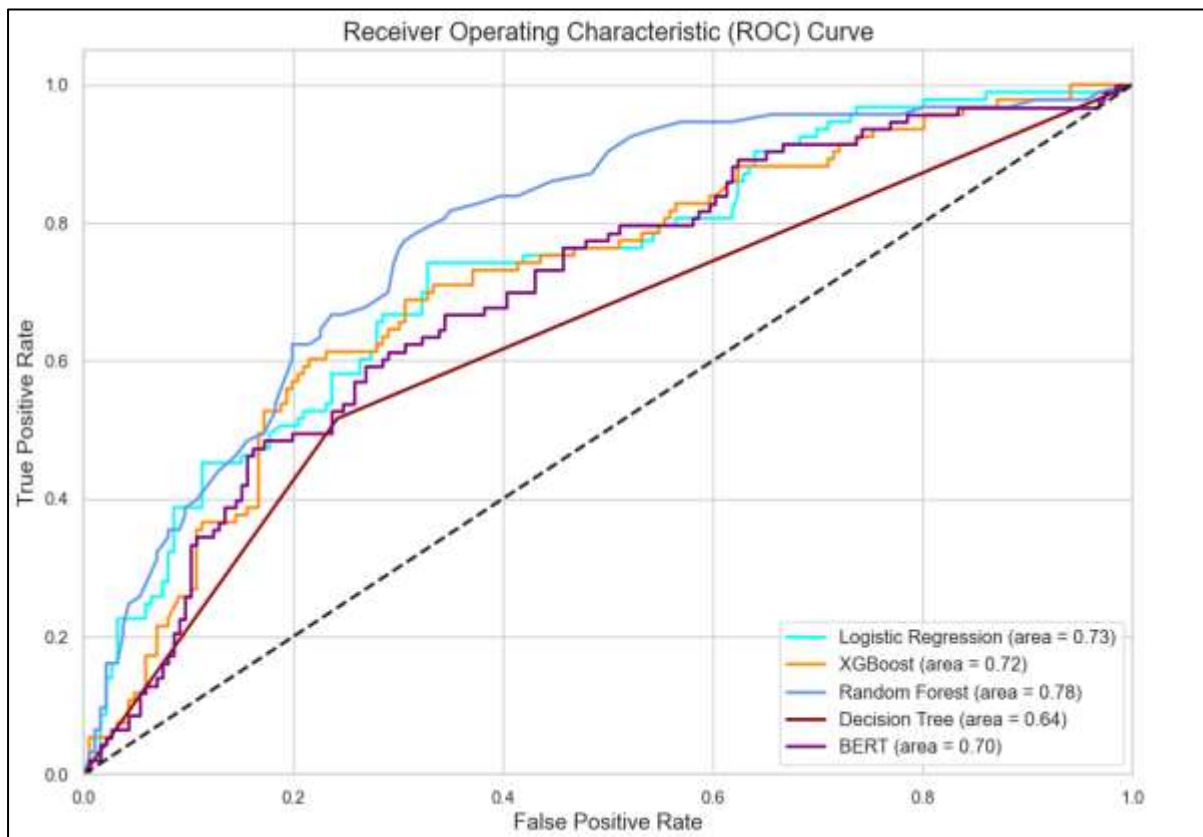


Figure 5.6 ROC Curve Analysis for Sentiment Analysis Models.

The ROC curve demonstrates the following performance for each model:

1. **Random Forest:** The Random Forest model achieves the highest AUC score of 0.78, indicating the best performance at distinguishing positive and negative sentiment in tweets.
2. **Logistic Regression:** Logistic Regression follows closely with an AUC of 0.73.
3. **XGBoost:** XGBoost achieves an AUC of 0.72.
4. **BERT:** The BERT model has an AUC score of 0.70.
5. **Decision Tree:** The model shows the lowest AUC of 0.64.

Based on the ROC curve analysis, the Random Forest model is the strongest at distinguishing between positive and negative sentiment in Twitter data. Logistic Regression, XGBoost, and BERT also perform well, while the Decision Tree model is the least effective in this task.

5.4 Comparative Analysis

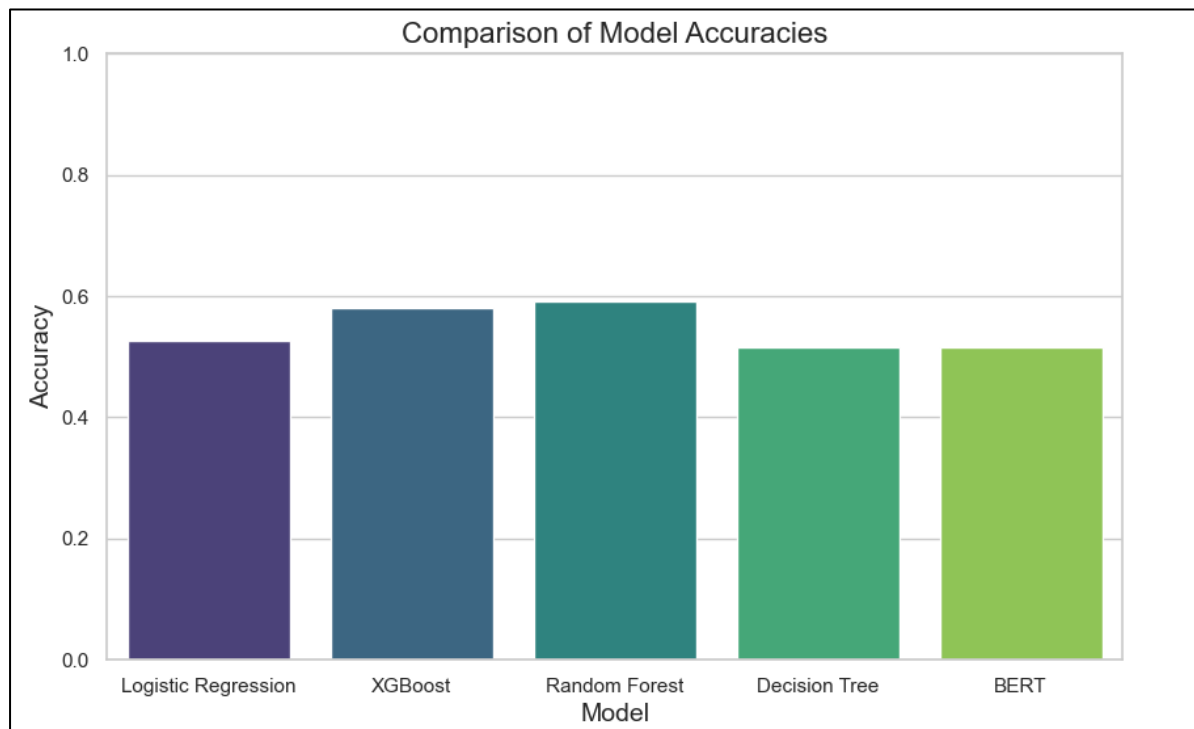


Figure 5.7: Comparative Analysis of Model Accuracies

Among the five models tested, Random Forest demonstrated the highest accuracy at 59.14%, followed by XGBoost with an accuracy of 58.06%. Logistic Regression, BERT, and Decision Tree had similar performances, with accuracies of around 52%.

The disparity in performance across models can be attributed to their inherent design and the specific characteristics of the dataset. While BERT, a transformer-based model, is known for handling large-scale NLP tasks, it might require more data and fine-tuning to outperform simpler models like Random Forest and XGBoost on this specific sentiment analysis task.

5.4.1 Validation Analysis for Models (User test Analysis)

After analyzing the sentiment predictions for five sample texts, it is evident that both the BERT and Logistic Regression models better align with the expected sentiment labels than the XGBoost model. BERT and Logistic Regression correctly identified the sentiment in four cases, while XGBoost only aligned correctly in one instance. Furthermore, XGBoost consistently showed higher confidence in its predictions (with probability scores around 0.73-0.96), even when the predictions were incorrect. In contrast, BERT and Logistic Regression exhibited more balanced probability scores, reflecting a more cautious approach in their sentiment classifications. This suggests that BERT and Logistic Regression are more reliable

for sentiment analysis tasks in this context, while XGBoost may require further tuning to improve its accuracy.

5.5 Validation Process for Sentiment Analysis Models (User Test)

Here, the validation process is used to evaluate the performance of 3 sentiment analysis models on sample text input with their expected sentiment results. The texts ranged from electronics reviews to comments on household items, with sentiments classified as Positive, Negative, or Neutral. The objective was to determine how well each model (BERT, XGBoost, Logistic Regression) predicted the sentiment of given text inputs compared to the expected results.

5.6 Conclusion

The study compared the performance of five machine learning models in sentiment analysis of tweets related to the #BlackLivesMatter political movement among women. The Random Forest model emerged as the most effective, followed closely by XGBoost. Logistic Regression, BERT, and Decision Tree showed moderate performance. Improvement recommendations include hyperparameter tuning, increasing training data, and exploring advanced techniques. The validation process indicated that BERT and Logistic Regression models are more reliable for sentiment analysis in this context, while XGBoost requires further refinement.

6 ANALYSIS OF RESULTS

6.1 Introduction.

This chapter offers an analysis of the experiment's results. The chapter will begin with a summary of key findings from the exploratory analysis and modelling phases. Next, the chapter will cover the implications of the research results on the state of the art and the market. Finally, the chapter will mention the factors that limited this research.

6.2 Key Findings

6.2.1 *Key findings from the EDA*

The dataset used in the research was imbalanced. The positive, one of three labels, had more than half of the instances in the dataset. However, they show the general mood of the public towards the campaign. Additionally, positive tweets had more variance than positive or negative tweets, indicating support and passion. Most tweet activity occurs in the latter half of the day, showing that most people in the movement are working citizens. The majority of the tweets in the datasets had very few likes, retweets, and shares, suggesting few accounts control most of the discourse.

6.2.2 *Key Findings from the Modelling Phase*

The baseline model was the logistic regression. The model scored around 50% accuracy, with 0.36 recall. A low performance from the baseline model shows the task is relatively complex. XGBoost is also a popular model known for high accuracy and precision. However, the model achieved 58% accuracy and 0.58 recall. These are poor results for the XGBoost model, which further reiterates the complexity of the task.

The best algorithm in the experiment was the random forest, with 59% accuracy and 51% recall. These results imply that the model is operating at half the desired effectiveness. It was also peculiar that the random forest performed better than XGBoost which has 58.06% accuracy and the complex BERT deep learning model which scored 52%. Further research will investigate the phenomenon and raise the models' performance.

6.3 Implications of the research.

Twitter is a great place to launch social media campaigns. Successful implementation of this research would assist professionals carrying out campaigns with an effective sentiment analyser. The model can help practitioners rate their campaigns' success and how the public reacts to them.

The model can also be extended for deeper analysis. This model considered only the tweets such as likes and retweets. However, the data could also feature more fine-grained information such as age and gender. This can be used by marketers and authorities to target demographics during the campaign and gather feedback on how the demographic is reacting to the campaign.

The political nature of the research data also implies the model could be used by government authorities to track sentiments on political movements and policies. They can use the information to ensure security and collect feedback on policies. However, allowing governments to track citizens leads to issues about civil liberties.

6.4 Limitations of the research.

6.4.1 *Informal Language:*

Tweets often contain informal language, slang, abbreviations, and emojis, which can complicate sentiment analysis. While BERT is designed to handle diverse text, it may still face difficulties with the unique characteristics of Twitter language.

6.4.2 *Sarcasm and Irony:*

Tweets often contain sarcasm or irony, which are difficult for models to interpret correctly. Misclassification of such tweets can lower overall performance.

6.4.3 *Subjectivity*

Sentiment analysis inherently involves subjective interpretation. Tweets expressing nuanced or mixed sentiments can be challenging for the model to classify accurately.

6.5 Conclusion

This chapter has analysed the results of the research. It began with a summary of key findings from the exploratory phase and the modelling phase. These gave insights into the data and model results. The next section discusses the implications of the research for the market. It covered points such as its use in marketing and government campaigns. Finally the research covered the limitations that contributed to the unexpected performance.

7 CONCLUSIONS

7.1 Introduction

This chapter aims to give the final thoughts on the research. The chapter will begin with a brief summary of the research. Next, an analysis of the achievement of objectives will be conducted. Finally, this chapter will give recommendations that could help researchers overcome limitations in this research.

7.2 Conclusion

This research aimed to create an effective sentiment analysis model to analyze sentiments on Twitter posts related to the #BlackLivesMatter political movement among women. This information could be helpful to social scientists, activists, and policymakers interested in understanding public sentiment and engagement with the movement. The research utilized tweets obtained through web scraping employing Selenium, a web automation tool. The tweets were pre-processed to remove noise, slang, and irregular text formats. Preliminary analysis provided important insights, such as the frequency and patterns of tweet engagement metrics. The research included two rounds of experimentation: the first with traditional machine learning algorithms (Logistic Regression, XGBoost, Random Forest, and Decision Tree) and the second with the transformer-based model BERT. Overall, BERT demonstrated promising performance, although there was room for improvement given the unique challenges posed by social media language.

7.3 Achievement of Objectives

7.3.1 Objective 1: Review and Critically Evaluate Existing Literature on Sentiment Analysis Methods

This objective was achieved by conducting a comprehensive literature review highlighting the strengths and limitations of various sentiment analysis techniques, including traditional machine learning algorithms and advanced models like BERT.

7.3.2 Objective 2: Data Collection and Pre-processing

This objective was met by collecting a diverse dataset of tweets related to the #BlackLivesMatter movement and implementing thorough pre-processing steps to ensure the data was suitable for model training.

7.3.3 Objective 3: Visualization of Tweet Engagement Metrics

Over time, the visualization of tweet engagement metrics identified vital trends and periods of heightened activity, achieving this objective and providing insights into the dynamic nature of social media discussions.

7.3.4 Objective 4: Design, Develop, and Implement Machine Learning Models

This objective was accomplished by designing and implementing several machine learning models, including Logistic Regression, XGBoost, Random Forest, Decision Tree, and BERT. The models were trained and tested on the pre-processed dataset, with BERT demonstrating the most promising results.

7.3.5 Objective 5: Evaluate ML Model Performance

The evaluation of model performance using standard metrics (accuracy, precision, recall, and F1 score) was successfully completed, highlighting each model's strengths and weaknesses. BERT showed the best balance between precision and recall, although further optimization is needed.

7.4 Recommendations

This research achieved notable results but also identified areas for improvement and further exploration. The following recommendations are provided to improve or expand upon this research:

7.4.1 Data Diversification and Volume

Future research should use different data sources to evaluate the performance of sentiment analysis models on varied difficulty levels. This includes incorporating tweets from various periods, regions, and contexts to capture the nuances of social media language.

A larger dataset should expose the models to a broader range of scenarios. Collaboration with organizations that can access extensive and reliable data is recommended.

7.4.2 Model Optimization

While this research used the Adam optimizer, future work should explore and justify the choice of different optimization algorithms. Researchers should experiment with various optimizers to identify the most effective one for sentiment analysis in social media contexts.

7.4.3 *Handling Informal Language and Emojis*

Future models should incorporate techniques to handle better informal language, slang, and emojis, which are prevalent in social media posts. This includes utilizing advanced pre-processing methods and integrating models designed to understand such content.

7.5 Conclusion

This chapter has provided the final thoughts on this research. It began with a brief summary of the research. Next, the chapter proved how the study achieved each of the objectives. Finally, the chapter described recommendations that could help other researchers who wish to improve or replicate this project.

8 FUTURE WORK

8.1 Introduction

This chapter aims to mention the plans that this research has for the future. These plans are meant to avoid some of the limitations that the research faced. The following sections covers each of the identified areas.

8.2 Incorporating Emojis in Sentiment Analysis

Future research should investigate the role of emojis in sentiment analysis. Emojis are integral to online communication and can provide valuable sentiment information. Incorporating emoji analysis could improve the accuracy of sentiment classification.

8.3 Optimization Techniques

Further work should focus on researching and selecting the most effective optimization algorithms for transformer models like BERT. This includes comparing the performance of different optimizers and fine-tuning hyperparameters to enhance model accuracy and reliability.

8.4 Multimodal Sentiment Analysis

Future research could explore multimodal sentiment analysis by combining text, images, and videos. This approach can provide a more comprehensive understanding of sentiments expressed in social media posts, offering more profound insights into public opinion.

8.5 Longitudinal Analysis

Conducting a longitudinal analysis of sentiment trends could reveal how public opinion evolves in response to events related to the #BlackLivesMatter movement. This can provide valuable insights for activists, policymakers, and social scientists.

9 REFERENCES

- Ahaotu, J.O. and Oshamo, O.A., 2023. A Multimodal Discourse Analysis of Selected Social Media Posts on the# BlackLivesMatter Protest. *Journal of Pragmatics and Discourse Analysis*, 2(1), pp.25-35.
- Alfano, M., Reimann, R., Quintana, I.O., Chan, A., Cheong, M. and Klein, C., 2023. The affiliative use of emoji and hashtags in the Black Lives Matter movement in Twitter. *Social Science Computer Review*, 41(6), pp.2145-2160.
- APAMPA, O., 2020. Social media movements: A data-driven analysis of gender issues, social advocacy, climate change and environmental communication on selected online social media platforms.
- Badaoui, S., 2020. Black Lives Matter: A new perspective from Twitter data mining. *Policy Center for the New South*, pp.2021-01.
- Bowman Williams, J., Mezey, N. and Singh, L., 2021. # BlackLivesMatter—Getting from contemporary social movements to structural change. *California Law Review Online*, 12.
- Brünker, F., Wischnewski, M., Mirbabaie, M. and Meinert, J., 2020. The role of social media during social movements—observations from the# metoo debate on Twitter.
- Chan, A., Ferreira, M. and Alfano, M., 2023. Reactionary attitudes: Strawson, Twitter, and the Black Lives Matter movement. In *Experiments in Moral and Political Philosophy* (pp. 331-355). Routledge.
- Crowder, C., 2023. When# BlackLivesMatter at the Women’s March: a study of the emotional influence of racial appeals on Instagram. *Politics, Groups, and Identities*, 11(1), pp.55-73.
- Destine, S., 2022. From a hashtag to a movement: Black women movement actors’ challenges to leading a radical movement in a “Postracial America”. *Humanity & Society*, 46(1), pp.28-51.
- Devlin, J., Chang, M.W., Lee, K. and Toutanova, K., 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

Dunklin, M. and Jennings, P., 2022. “Where’s the Outrage??”: An Analysis of# BlackLivesMatter and# BlackTransLivesMatter Twitter Counterpublics. *Journalism & Mass Communication Quarterly*, 99(3), pp.763-783.

Field, A., Park, C.Y., Theophilo, A., Watson-Daniels, J. and Tsvetkov, Y., 2022. An analysis of emotions and the prominence of positivity in# BlackLivesMatter tweets. *Proceedings of the National Academy of Sciences*, 119(35), p.e2205767119.

Grimes, A., 2019. # BlackLivesMatter: How a hashtag maintains a movement: A content analysis of the Black Lives Matter hashtag on Twitter.

Guo, J. and Liu, S., 2022. From# BlackLivesMatter to# StopAsianHate: Examining network agenda-setting effects of hashtag activism on Twitter. *Social Media+ Society*, 8(4), p.20563051221146182.

Haffner, M., 2019. A place-based analysis of# BlackLivesMatter and counter-protest content on Twitter. *GeoJournal*, 84(5), pp.1257-1280.

Hotz, N., 2024. What is CRISP DM?. *Data Science Process Alliance*.
<https://www.datascience-pm.com/crisp-dm-2/>

Hswen, Y., Huerta, D.T., Le-Compte, C., Hawkins, J.B. and Brownstein, J.S., 2021. A 10-year social media analysis exploring hospital online support of Black lives Matter and the Black community. *JAMA Network Open*, 4(10), pp.e2126714-e2126714.

Jackson, J., 2019. # BlackLivesMatter: A Mixed Methods Exploratory Study of Tweets and Individuals Participating in the Black Lives Matter Movement (Doctoral dissertation, Pepperdine University).

Jiang, K. and Xu, Q., 2023. Analyzing the dynamics of social media texts using coherency network analysis: a case study of the tweets with the co-hashtags of# BlackLivesMatter and# StopAsianHate. *Frontiers in Research Metrics and Analytics*, 8.

Jones, K., Nurse, J.R. and Li, S., 2022, May. Out of the shadows: analyzing anonymous’ Twitter resurgence during the 2020 black lives matter protests. In *Proceedings of the International AAAI Conference on Web and Social Media* (Vol. 16, pp. 417-428).

- Klein, C., Reimann, R., Quintana, I.O., Cheong, M., Ferreira, M. and Alfano, M., 2022. Attention and counter-framing in the Black Lives Matter movement on Twitter. *Humanities and Social Sciences Communications*, 9(1), pp.1-12.
- Lampinen, A., 2020. *Tweeting for change: how Twitter users practice hashtag activism through #BlackLivesMatter and #MeToo* (Master's thesis, A. Lampinen).
- LaValley, M.P., 2008. Logistic regression. *Circulation*, 117(18), pp.2395-2399.
- Leach, C.W. and Teixeira, C.P., 2022. Understanding sentiment toward “black lives matter”. *Social Issues and Policy Review*, 16(1), pp.3-32.
- Li, M., Turki, N., Izaguirre, C.R., DeMahy, C., Thibodeaux, B.L. and Gage, T., 2021. Twitter as a tool for social movement: An analysis of feminist activism on social media communities. *Journal of community psychology*, 49(3), pp.854-868.
- Nartey, M., 2022. Centering marginalized voices: A discourse analytic study of the Black Lives Matter movement on Twitter. *Critical Discourse Studies*, 19(5), pp.523-538.
- Saunders, M.N., Lewis, P., Thornhill, A. and Bristow, A., 2015. Understanding research philosophy and approaches to theory development.
- Schuetze, S., 2023. Insights into a New Social Movement: Utilizing Machine Learning to Classify and Analyze #BlackLivesMatter Twitter Advocacy.
- Syed, R. and Silva, L., 2023. Social movement sustainability on social media: An analysis of the women’s March Movement on Twitter. *Journal of the Association for Information Systems*, 24(1), pp.249-293.
- Tanksley, T.C., 2019. *Race, education and #BlackLivesMatter: How social media activism shapes the educational experiences of Black college-age women*. University of California, Los Angeles.
- Tillery, A.B., 2019. What kind of movement is Black Lives Matter? The view from Twitter. *Journal of Race, Ethnicity, and Politics*, 4(2), pp.297-323.
- Tong, X., Li, Y., Li, J., Bei, R. and Zhang, L., 2022, July. What are People Talking about in #backlivesmatter and #stopasianhate? Exploring and Categorizing Twitter Topics Emerged in

Online Social Movements through the Latent Dirichlet Allocation Model. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 723-738).

Toraman, C., Şahinuç, F. and Yilmaz, E.H., 2022, June. Blacklivesmatter 2020: an analysis of deleted and suspended users in Twitter. In *Proceedings of the 14th ACM Web Science Conference 2022* (pp. 290-295).

van Haperen, S., Uitermark, J. and Nicholls, W., 2023. The Swarm versus the Grassroots: Places and networks of supporters and opponents of Black Lives Matter on Twitter. *Social Movement Studies*, 22(2), pp.171-189.

Vitali, M.M., Presotto, G.C., Gizzi, F., Gomes, M.D.A. and Giacomozzi, A.I., 2021. #BlackLivesMatter: A study of social representations from Twitter. *Community Psychology in Global Perspective*, 8(1), pp.1-19.

Waterhouse, J.C., 2022. #BlackLivesMatter: critical political implications of Twitter discourse in the wake of George Floyd.

Wilkins, D.J., Livingstone, A.G. and Levine, M., 2019. Whose tweets? The rhetorical functions of social media use in developing the Black Lives Matter movement. *British Journal of Social Psychology*, 58(4), pp.786-805.

10 APPENDIX

10.1 Appendix A: Setting up Login

Setup the login

```
try:
    username = WebDriverWait(driver, 10).until(
        EC.presence_of_element_located((By.XPATH, "//input[@name='text']"))
    )
    username.send_keys(username_str)
    next_button = driver.find_element(By.XPATH, "//span[contains(text(),'Next')]")
    next_button.click()
except (NoSuchElementException, TimeoutException, WebDriverException) as e:
    print(f"Error during login: {e}")
    driver.quit()

try:
    password = WebDriverWait(driver, 10).until(
        EC.presence_of_element_located((By.XPATH, "//input[@name='password']"))
    )
    password.send_keys(password_str)
    log_in = driver.find_element(By.XPATH, "//span[contains(text(),'Log in')]")
    log_in.click()
except (NoSuchElementException, TimeoutException, WebDriverException) as e:
    print(f"Error during login: {e}")
    driver.quit()
```

10.2 Appendix B: Obtaining Tweets from the #BlackLivesMatter hashtag

Search for the hashtag #BlackLivesMatter

```
try:
    search_box = WebDriverWait(driver, 10).until(
        EC.presence_of_element_located((By.XPATH, "//input[@data-testid='SearchBox_Search_Input']"))
    )
    search_box.send_keys("Black Lives Matter")
    search_box.send_keys(Keys.ENTER)
except (NoSuchElementException, TimeoutException, WebDriverException) as e:
    print(f"Error during search: {e}")
    driver.quit()
```

Click on the "People" tab

```
try:
    people_tab = WebDriverWait(driver, 10).until(
        EC.element_to_be_clickable((By.XPATH, "//span[contains(text(),'People')]"))
    )
    people_tab.click()
except (NoSuchElementException, TimeoutException, WebDriverException) as e:
```

```

    print(f"Error during switching to people: {e}")
driver.quit()

```

10.3 Appendix C: Collecting the tweets from the hashtag.

Click on the first profile in the "People" tab

try:

```

    first_profile = WebDriverWait(driver, 10).until(
        EC.element_to_be_clickable((By.XPATH, "//*[@id='react-
root']/div/div/div[2]/main/div/div/div/div/div/div[3]/section/div/div/div[1]/div/div/button/div/div[2]/div[1]/
div[1]/div/div[1]/a/div/div[1]/span/span[1]"))
    )
    first_profile.click()
except (NoSuchElementException, TimeoutException, WebDriverException) as e:
    print(f"Error during profile selection: {e}")
driver.quit()

```

Initialize lists to store tweet data

```

UserTags = []
TimeStamps = []
Tweets = []
Replies = []
Retweets = []
Likes = []

```

Fetch tweets from the profile

try:

```

    sleep(10)
    last_height = driver.execute_script("return document.body.scrollHeight")
    while True:
        articles = driver.find_elements(By.XPATH, "//article[@data-testid='tweet']")
        for article in articles:
            try:
                UserTag = article.find_element(By.XPATH, "//*[@data-testid='User-Name']").text
                TimeStamp = article.find_element(By.XPATH, "//*[@time]").get_attribute('datetime')
                Tweet = article.find_element(By.XPATH, "//*[@data-testid='tweetText']").text
                Reply = article.find_element(By.XPATH, "//*[@button[@data-testid='reply']]").text
                Retweet = article.find_element(By.XPATH, "//*[@button[@data-testid='retweet']]").text
                Like = article.find_element(By.XPATH, "//*[@button[@data-testid='like']]").text

                UserTags.append(UserTag)
                TimeStamps.append(TimeStamp)
                Tweets.append(Tweet)
                Replies.append(Reply)
                Retweets.append(Retweet)
                Likes.append(Like)
            except NoSuchElementException as e:
                print(f"Error: {e}")

```

```

driver.execute_script('window.scrollTo(0,document.body.scrollHeight);')
sleep(5)
new_height = driver.execute_script("return document.body.scrollHeight")
if new_height == last_height: # Break if no new tweets are loaded
    break
last_height = new_height

# Optionally break after a certain number of tweets
if len(Tweets) > 4000: # Adjust as needed
    break

# Print the number of collected tweets
print(len(UserTags), len(TimeStamps), len(Tweets), len(Replies), len(Retweets), len(Likes))

# Save data to a DataFrame and CSV file
df = pd.DataFrame(zip(UserTags, TimeStamps, Tweets, Replies, Retweets, Likes),
                  columns=['UserTags', 'TimeStamps', 'Tweets', 'Replies', 'Retweets', 'Likes'])

df.head()

# Save the DataFrame to a CSV file in the current working directory
df.to_csv("tweets_live.csv", index=False)

print("Tweets have been saved to 'tweets_live.csv'")

except WebDriverException as e:
    print(f"Error during tweet fetching: {e}")
finally:
    # Close the driver
    driver.quit()

```

10.4 Appendix D: Data cleaning and text preprocessing

```

df['TimeStamps'] = pd.to_datetime(df['TimeStamps'])

def clean_tweet(tweet):
    if isinstance(tweet, str):
        tweet = re.sub(r'http\S+', "", tweet) # Remove URLs
        tweet = re.sub(r'@\S+', "", tweet)    # Remove mentions
        tweet = re.sub(r'#\S+', "", tweet)    # Remove hashtags
        tweet = re.sub(r'[^A-Za-z0-9\s]', "", tweet) # Remove special characters
        tweet = tweet.lower() # Convert to lowercase
    else:
        tweet = "" # If the tweet is not a string, convert to an empty string
    return tweet

df['Cleaned_Tweets'] = df['Tweets'].apply(clean_tweet)

```

```

# Process the numeric columns
def process_k_values(value):
    value = str(value)
    if 'K' in value:
        return float(value.replace('K', '')) * 1000
    return float(value)

df['Replies'] = df['Replies'].apply(process_k_values)
df['Retweets'] = df['Retweets'].apply(process_k_values)
df['Likes'] = df['Likes'].apply(process_k_values)

```

10.5 Appendix E: Creating and visualizing sentiment labels

```

# Create sentiment labels
df['Sentiment_Label'] = df['Sentiment'].apply(lambda x: 'Positive' if x > 0 else ('Negative' if x < 0 else 'Neutral'))

# Prepare data for the treemap
sentiment_counts = df['Sentiment_Label'].value_counts().reset_index()
sentiment_counts.columns = ['Sentiment', 'Count']

# Calculate percentages
total_count = sentiment_counts['Count'].sum()
sentiment_counts['Percentage'] = (sentiment_counts['Count'] / total_count) * 100
sentiment_counts['Label'] = sentiment_counts.apply(lambda x: f"{x['Sentiment']}\n{x['Count']}\n({x['Percentage']:.2f}%)", axis=1)

# Plot the treemap with percentages
plt.figure(figsize=(12, 8))
colors = ['green', 'red', 'blue']
squarify.plot(sizes=sentiment_counts['Count'], label=sentiment_counts['Label'], color=colors, alpha=.8)
plt.title('Sentiment Distribution of #BlackLivesMatter Tweets')
plt.axis('off')
plt.show()

```

10.6 Appendix F: Logistic Regression Modelling

```

# Split the data
X = df['Cleaned_Tweets']
y = df['Sentiment_Label']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Vectorize the text data
tfidf = TfidfVectorizer(max_features=10000, ngram_range=(1, 2))
X_train_tfidf = tfidf.fit_transform(X_train)
X_test_tfidf = tfidf.transform(X_test)

```

```
# Train the Logistic Regression model
log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train_tfidf, y_train)

# Evaluate the model
y_pred_log_reg = log_reg.predict(X_test_tfidf)
print("Logistic Regression:")
print(classification_report(y_test, y_pred_log_reg))
print("Accuracy:", accuracy_score(y_test, y_pred_log_reg))
```


10.7 Appendix G: XGBOOST Modelling

Encode sentiment labels to numeric values

```
label_encoder = LabelEncoder()
y_train_encoded = label_encoder.fit_transform(y_train)
y_test_encoded = label_encoder.transform(y_test)
```

Train the XGBoost model

```
xgb_model = xgb.XGBClassifier(eval_metric='mlogloss')
xgb_model.fit(X_train_tfidf, y_train_encoded)
```

Evaluate the model

```
y_pred_xgb = xgb_model.predict(X_test_tfidf)
print("XGBoost:")
print(classification_report(y_test_encoded, y_pred_xgb, target_names=label_encoder.classes_))
print("Accuracy:", accuracy_score(y_test_encoded, y_pred_xgb))
```

10.8 Appendix H: BERT Modelling

Encode the labels

```
label_encoder = LabelEncoder()
df['Sentiment_Label'] = label_encoder.fit_transform(df['Sentiment_Label'])
y = df['Sentiment_Label']
```

Tokenize the data

```
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
```

```
def encode_data(texts, labels, max_length=128):
```

```
    input_ids = []
```

```
    attention_masks = []
```

```
    for text in texts:
```

```
        encoded = tokenizer.encode_plus(
```

```
            text,
```

```
            add_special_tokens=True,
```

```
            max_length=max_length,
```

```
            truncation=True,
```

```
            padding='max_length',
```

```
            return_attention_mask=True,
```

```
            return_tensors='pt'
```

```
        )
```

```
        input_ids.append(encoded['input_ids'])
```

```
        attention_masks.append(encoded['attention_mask'])
```

```
    return torch.cat(input_ids, dim=0), torch.cat(attention_masks, dim=0), torch.tensor(labels,
dtype=torch.long)
```

Convert texts to list and labels to NumPy array

```

X_train_list = X_train.tolist()
X_test_list = X_test.tolist()
y_train_encoded = label_encoder.transform(y_train)
y_test_encoded = label_encoder.transform(y_test)

X_train_ids, X_train_masks, y_train_tensor = encode_data(X_train_list, y_train_encoded)
X_test_ids, X_test_masks, y_test_tensor = encode_data(X_test_list, y_test_encoded)

# Create a custom dataset
class CustomDataset(torch.utils.data.Dataset):
    def __init__(self, input_ids, attention_masks, labels):
        self.input_ids = input_ids
        self.attention_masks = attention_masks
        self.labels = labels

    def __len__(self):
        return len(self.labels)

    def __getitem__(self, idx):
        return {
            'input_ids': self.input_ids[idx],
            'attention_mask': self.attention_masks[idx],
            'labels': self.labels[idx]
        }

train_dataset = CustomDataset(X_train_ids, X_train_masks, y_train_tensor)
test_dataset = CustomDataset(X_test_ids, X_test_masks, y_test_tensor)

# Define the BERT model
model = BertForSequenceClassification.from_pretrained('bert-base-uncased',
num_labels=len(label_encoder.classes_))

# Define training arguments
training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=3,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    warmup_steps=500,
    weight_decay=0.01,
    logging_dir='./logs',
    logging_steps=10,
    evaluation_strategy='epoch'
)

# Train the model
trainer = Trainer(
    model=model,

```

```
args=training_args,  
train_dataset=train_dataset,  
eval_dataset=test_dataset  
)  
  
trainer.train()  
  
# Evaluate the model  
predictions = trainer.predict(test_dataset)  
preds = predictions.predictions.argmax(-1)  
print("BERT:")  
print(classification_report(y_test_encoded, preds, target_names=label_encoder.classes_))  
print("Accuracy:", accuracy_score(y_test_encoded, preds))
```

10.9 Appendix I: Random Forest Modelling

Split the data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Vectorize the text data

```
tfidf = TfidfVectorizer(max_features=10000, ngram_range=(1, 2))
```

```
X_train_tfidf = tfidf.fit_transform(X_train)
```

```
X_test_tfidf = tfidf.transform(X_test)
```

Train the Random Forest model

```
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
```

```
rf_model.fit(X_train_tfidf, y_train)
```

Evaluate the Random Forest model

```
y_pred_rf = rf_model.predict(X_test_tfidf)
```

```
print("Random Forest:")
```

```
print(classification_report(y_test, y_pred_rf))
```

```
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
```

Train the Decision Tree model

```
dt_model = DecisionTreeClassifier(random_state=42)
```

```
dt_model.fit(X_train_tfidf, y_train)
```

Evaluate the Decision Tree model

```
y_pred_dt = dt_model.predict(X_test_tfidf)
```

```
print("Decision Tree:")
```

```
print(classification_report(y_test, y_pred_dt))
```

```
print("Accuracy:", accuracy_score(y_test, y_pred_dt))
```

10.10 Appendix J: Ethics Application Form

Faculty of Science, Business & Enterprise Science & Engineering Research Project Form - Student

*Please type your responses below and email to SciEng-REC@chester.ac.uk
Handwritten copies will not be accepted.*

Your Name:	DHARANI MANIKANDAN
Your student number:	2323200
Email Address:	2323200@Chester.ac.uk
Programme of Study:	M.Sc. Data Science
Name of Principal Supervisor:	Mohammed Saeed
Title of Research Project:	Sentiment Analysis of Social Media Post on #BlackLivesMatter Political Movement among women using Twitter platform
Start date of project: March 8, 2024	Anticipated end date of project: October 3, 2024

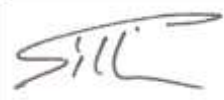
Please explain why you want to carry out this research and what your proposed method will be (no more than 100 words):

The aim of the dissertation is to conduct a comprehensive sentiment analysis of Twitter social media posts, utilizing NLP and ML techniques, to gain insights into the emotional tone of online conversations regarding political movements (#BlackLivesMatter).

Please respond to the following questions:

Question	Response
Will your research be based on reviewing existing literature?	Entirely ☐ partly ☒ Not at all ☐ Not sure at this stage ☐
Will your research involve mathematical work?	Yes ☐ No ☐ Not sure at this stage ☒
Are you planning to develop or test any computer system or piece of hardware or software?	Yes ☐ No ☒ Not sure at this stage ☐
Will your research involve you carrying out testing using an isolated network or virtual machine?	Yes ☐ No ☒ Not sure at this stage ☐
Are you intending to use research data available from an online source?	Yes ☒ No ☐ Not sure at this stage ☐
If yes, have you checked that there are no copyright or data protection issues involved in you working with and reproducing this data?	Yes, I've checked and there are no issues ☒ Yes, I've checked and there are issues ☐ No, I haven't checked ☐
Will your research involve work carried out in a science laboratory?	Yes ☐ No ☒ Not sure at this stage ☐
Will your research involve fieldwork?	Yes ☐ No ☒ Not sure at this stage ☐
If you have answered 'yes' to either of these, have you carried out a Risk Assessment?	Yes ☐ No ☒ Not sure at this stage ☐
Risk assessment reference number:	
Will you need to liaise with the Laboratory Manager regarding any special requirements to be observed in addition to standard lab procedures and PPE?	Yes ☐ No ☒
Will your project involve you having contact with human participants, e.g. through	Yes ☐ No ☒ Not sure at this stage ☐

interviews, focus groups, data gathering via questionnaires, surveys on social media, etc.?	
Will your project involve you having contact with animals or animal tissues?	Yes ☐ No ☒ Not sure at this stage ☐
If you are not working directly with animals, or animal tissues, are you using research data about these which has collected by another person or organisation?	Yes ☐ No ☒ Not sure at this stage ☐
Is permission needed to use this data?	Yes, permission is needed and I have got permission ☐ No permission is required ☒ I haven't checked ☐
Does your project involve the NHS in any way?	Yes ☐ No ☒ Not sure at this stage ☐
Is your project likely to engage with the natural environment, e.g. by utilising samples collected from nature, producing hazardous chemical by-products, creating noise pollution, etc.?	Yes ☐ No ☒ Not sure at this stage ☐
Will your project, including data-gathering or collaborative activities, involve research outside of England?	Yes ☐ No ☒ Not sure at this stage ☐
If you are likely to travel outside of England to conduct research, where are you intending to go?	Please provide details:
Are you aware of any risks to you in travelling to the destinations named above?	Yes ☐ No ☒
Do you think this project might need ethical approval?	Yes ☒ No ☐
Have you discussed this with your supervisor?	Yes ☒ No ☐

Your signature:	M.Dharani Manikandan
Supervisor's Signature:	<i>MASaeed</i>
Comments from the Science & Engineering Research Ethics Committee	No ethical concerns identified.
Signature of the Chair of the Science & Engineering Research Ethics Committee:	
Date:	09/04/24

10.11 Appendix K: Data Set (Link)

[J112467_Dataset](#)